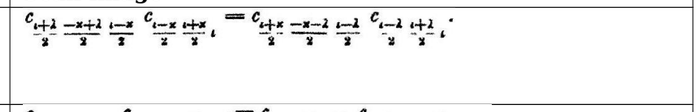
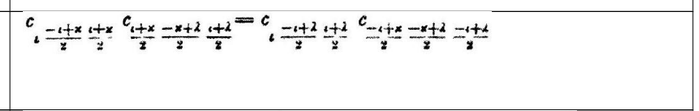
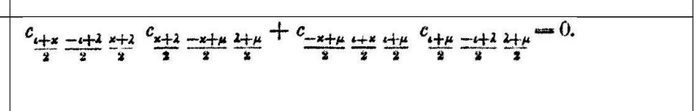
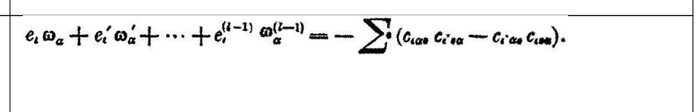
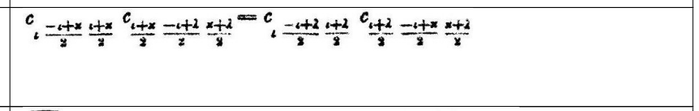
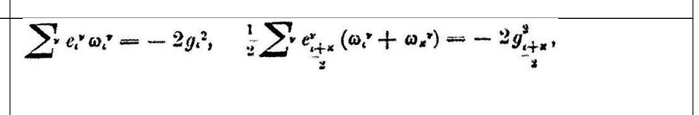
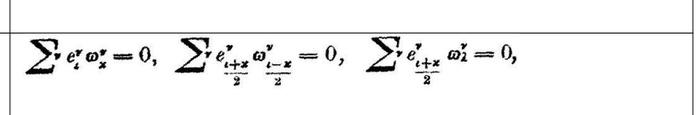
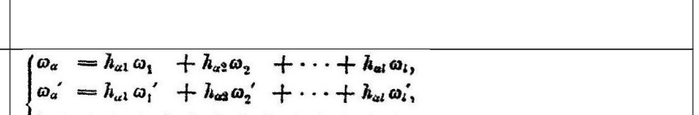
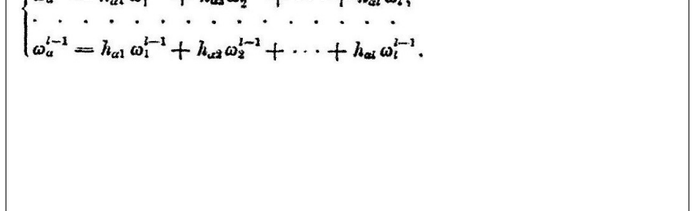
killing1889 — formula report

763 inline formulas (0 shown: the ones that did not render), 194 display equations, 0 tables, 0 unrecovered image regions.

Display equations

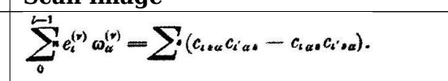
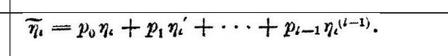
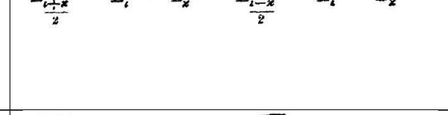
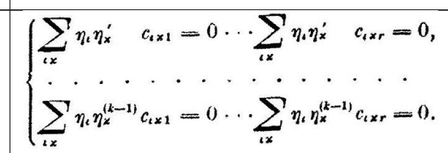
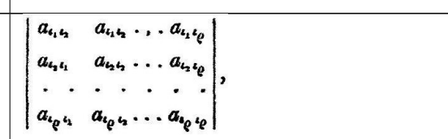
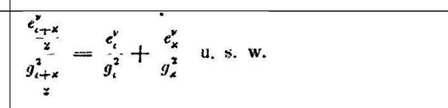
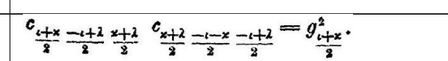
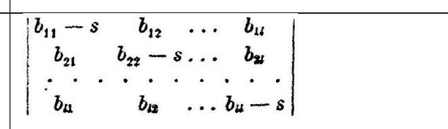
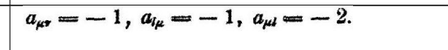
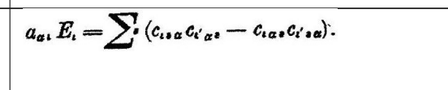
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
killing1889_ EQ0157 (9) **[conf 0.006]**	041	■ 0.006 ● X|+3	`c_{\frac{4+2}{2}-x+2 \frac{1-x}{2}} c_{\frac{4-x}{2}} \frac{4+x}{2}, \quad c_{4+x} \frac{-x-2}{2} \frac{1-2}{2} c_{4-2} \frac{4+2}{2}`	$c_{\frac{4+2}{2}-x+2\frac{1-x}{2}}c_{\frac{4-x}{2}}\frac{4+x}{2}, \quad c_{4+x}\frac{-x-2}{2}\frac{1-2}{2}c_{4-2}\frac{4+2}{2}$	
killing1889_ EQ0156 (8) **[conf 0.014]**	041	■ 0.014 ● W|-1	`{ }_{t}^{c} \frac{-4+x}{2} \frac{4 x}{2} c_{\frac{4 x}{2}} \frac{-x+2}{2} \frac{4+2}{2}=c_{3} \frac{-4+2}{2} \frac{4+2}{2}=c_{3} \frac{-4+2}{2}} \frac{-x+2}{2} \frac{4+2}{2}=c_{3} \frac{-4+2}{2} \frac{4+2}{2} c_{-4+\frac{x}{2}}=\frac{-x+2}{2}=\frac{4+4}{2}`	${}_{t}^{c}\frac{-4+x}{2}\frac{4x}{2}c_{\frac{4x}{2}}\frac{-x+2}{2}\frac{4+2}{2}=c_3\frac{-4+2}{2}\frac{4+2}{2}c_{-4+\frac{x}{2}}=\frac{-x+2}{2}=\frac{4+4}{2}$	
killing1889_ EQ0158 (10) **[conf 0.017]**	041	■ 0.017 ● C|-21	`c_{\frac{1+x}{2}}=\frac{-+\lambda}{2} \frac{x+2}{2} c_{\frac{x+2}{2}} \frac{-x+\mu}{2} \frac{2+\mu}{2}+c_{\frac{-x+\mu}{2}}^{\frac{4+x}{2}} \frac{4+\mu}{2} c_{\frac{4+\mu}{2}}=\frac{-4+2}{2} \frac{2+\mu}{2}=0 .`	$c_{\frac{1+x}{2}}=\frac{-+\lambda}{2}\frac{x+2}{2}c_{\frac{x+2}{2}}\frac{-x+\mu}{2}\frac{2+\mu}{2}+c_{\frac{-x+\mu}{2}}^{\frac{4+x}{2}}\frac{4+\mu}{2}c_{\frac{4+\mu}{2}}=\frac{-4+2}{2}\frac{2+\mu}{2}=0.$	
killing1889_ EQ0060 (6) **[conf 0.019]**	015	■ 0.019 ● C|-17	`e_{i} \omega_{\alpha}+e_{i}^{\prime} \omega_{\alpha}^{\prime}+\cdots+e_{i}^{(i-1)} \omega_{\alpha}^{(i-1)}=-\sum\left(c_{i a o} c_{i r \alpha}-c_{i \alpha} c_{i \alpha \beta}\right) .`	$e_i\omega_\alpha+e_i'\omega_\alpha'+\cdots+e_i^{(i-1)}\omega_\alpha^{(i-1)}=-\sum\left(c_{iao}c_{ir\alpha}-c_{i\alpha}c_{i\alpha\beta}\right).$	
killing1889_ EQ0155 (7) **[conf 0.036]**	041	■ 0.036 ● C|-17	`c_{t} \frac{-i+x}{2} \frac{i+x}{2} c_{\frac{1+x}{2}}=\frac{-i+2}{2} \frac{x+2}{2}=c_{1}=\frac{-4+2}{2} \frac{1+2}{2} c_{1+2}=\frac{4+x}{2} \frac{x+2}{x}`	$c_t\frac{-i+x}{2}\frac{i+x}{2}c_{\frac{1+x}{2}}=\frac{-i+2}{2}\frac{x+2}{2}=c_1=\frac{-4+2}{2}\frac{1+2}{2}c_{1+2}=\frac{4+x}{2}\frac{x+2}{x}$	
killing1889_ EQ0002 **[conf 0.046]**	002	■ 0.046 ● C|-9	`\sum \ell\left\{c_{x 2_{p}}\left(X_{p} X_{i}\right)+c_{2_{i \ell}}\left(X_{p} X_{x}\right)+c_{i \rho}\left(X_{\rho} X_{2}\right)\right\}=0,`	$\sum \ell\left\{c_{x2_p}\left(X_pX_i\right)+c_{2_{i\ell}}\left(X_pX_x\right)+c_{i\rho}\left(X_\rho X_2\right)\right\}=0,$	
killing1889_ EQ0165 **[conf 0.065]**	043	■ 0.065 ● X|+2	`\sum e_{i}^{v} \omega_{i}^{v}=-2 g_{i}^{2}, \quad \frac{1}{2} \sum v_{i+x}^{v}\left(\omega_{i}^{v}+\omega_{x}^{v} \right)=-2 g_{i+x}^{2},`	$\sum e_i^v\omega_i^v=-2g_i^2, \quad \frac{1}{2}\sum v_{i+x}^v\left(\omega_i^v+\omega_x^v\right)=-2g_{i+x}^2,$	
killing1889_ EQ0149 (3) **[conf 0.067]**	040	■ 0.067 ● X|+2	`\sum e_{i}^{r} \omega_{x}^{v}=0, \quad \sum e_{\frac{i+x}{2}}^{y} \omega_{\frac{i-x}{2}}^{v}=0, \quad \sum_{\frac{t+x}{2}}^{y} \omega_{z}^{v}=0,`	$\sum e_i^r\omega_x^v=0, \quad \sum e_{\frac{i+x}{2}}^y\omega_{\frac{i-x}{2}}^v=0, \quad \sum_{\frac{t+x}{2}}^{y}\omega_z^v=0,$	
killing1889_ EQ0071 (11) **[conf 0.083]**	017	■ 0.083 ● X|+5	`\left\{\begin{array}{l} \omega_{\alpha}=h_{\alpha 1} \omega_{1}+h_{\alpha 2} \omega_{2}+\cdots+h_{\alpha 1} \omega_{1}, \\ \omega_{\alpha}=h_{\alpha 1} \omega_{1}^{\prime}+h_{\alpha \alpha} \omega_{2}^{\prime}+\cdots+h_{\alpha 1} \omega_{2}^{\prime}, \\ \omega_{\alpha}^{\prime-1}=h_{\alpha 1} \omega_{1}^{\prime-1}+h_{\alpha 2} \omega_{2}^{\prime-1}+\cdots+h_{\alpha i} \omega_{l}^{\prime-1} . \end{array}\right.`	$\begin{cases}\omega_\alpha=h_{\alpha1}\omega_1+h_{\alpha2}\omega_2+\cdots+h_{\alpha i}\omega_i,\\ \omega_\alpha'=h_{\alpha1}\omega_1'+h_{\alpha2}\omega_2'+\cdots+h_{\alpha i}\omega_i',\\ \cdots\cdots\cdots\cdots\cdots\cdots\\ \omega_\alpha^{i-1}=h_{\alpha1}\omega_1^{i-1}+h_{\alpha2}\omega_2^{i-1}+\cdots+h_{\alpha i}\omega_i^{i-1}.\end{cases}$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix's reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● X crop overran its region ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

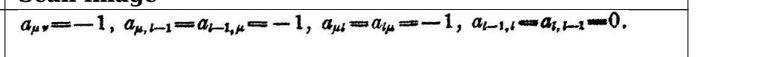
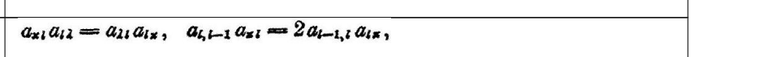
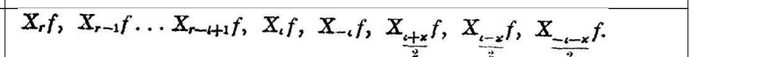
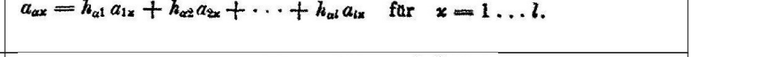
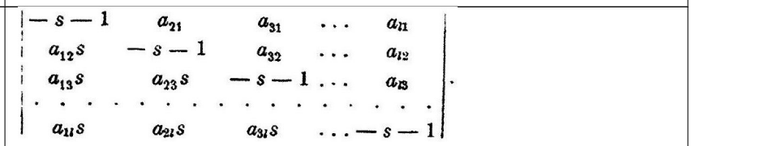
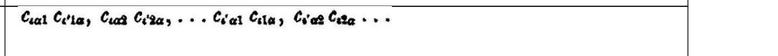
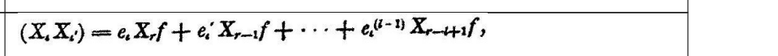
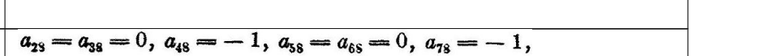
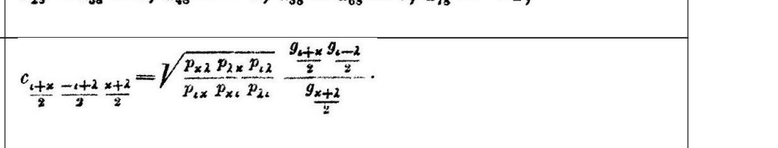
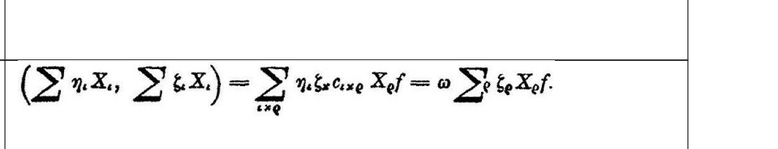
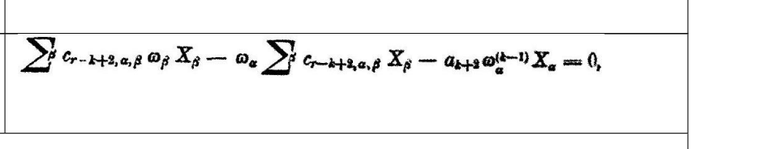
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
killing1889_EQ0135 (2) **[conf 0.099]**	033	■ 0.099 ●	`\sum_{0}^{L-1} e_{i}^{(\gamma)} \omega_{\alpha}^{(\gamma)}=\sum\left(c_{i+\alpha} c_{i \alpha}-c_{i \alpha} c_{i^{\prime}, \alpha}\right) .`	$\sum_{0}^{L-1} e_i^{(\gamma)}\omega_\alpha^{(\gamma)} = \sum\left(c_{i+\alpha}c_{i\alpha} - c_{i\alpha}c_{i',\alpha}\right).$	
killing1889_EQ0046	012	■ 0.114 ● C|-4	`\tilde{\eta}_{\iota}=p_{0} \eta_{\iota}+p_{1} \eta_{i}^{\prime}+\cdots+p_{i-1} \eta_{i}{ }^{(i-1)} .`	$\tilde{\eta}_\iota = p_0\eta_\iota + p_1\eta_i' + \cdots + p_{i-1}\eta_i^{(i-1)}.$	
killing1889_EQ0150 (4)	040	■ 0.119 ● X|+2	`\frac{\frac{e_{t+x}^{y}}{2}}{\frac{E_{t+x}^{y}}{2}}=\frac{e_{t}^{y}}{E_{t}}+\frac{e_{x}^{y}}{E_{x}}, \quad \frac{\frac{e_{t-x}^{y}}{2}}{\frac{E_{t-x}}{2}}=\frac{e_{t}^{y}}{E_{t}}-\frac{e_{x}^{y}}{E_{x}} .`	$\frac{\frac{e^y_{t+x}}{2}}{\frac{E^y_{t+x}}{2}} = \frac{e^y_t}{E_t} + \frac{e^y_x}{E_x}, \quad \frac{\frac{e^y_{t-x}}{2}}{\frac{E_{t-x}}{2}} = \frac{e^y_t}{E_t} - \frac{e^y_x}{E_x}.$	
killing1889_EQ0039 (2)	011	■ 0.133 ● X|+7	`\left\{\begin{array}{l} \sum_{\iota x} \eta_{\iota} \eta_{x}^{\prime} \quad c_{i x 1}=0 \cdots \sum_{\iota x} \eta_{i} \eta_{x}^{\prime} \quad c_{i x r}=0, \\ \sum_{\iota x} \eta_{\iota} \eta_{x}^{(k-1)} c_{t x 1}=0 \cdots \sum_{\iota x} \eta_{i} \eta_{x}^{(k-1)} c_{c x r}=0 . \end{array}\right.`	$\begin{cases} \sum_{\iota x} \eta_\iota \eta'_x \quad c_{ix1} = 0 \cdots \sum_{\iota x} \eta_i \eta'_x \quad c_{ixr} = 0, \\ \sum_{\iota x} \eta_\iota \eta_x^{(k-1)} c_{tx1} = 0 \cdots \sum_{\iota x} \eta_i \eta_x^{(k-1)} c_{cxr} = 0. \end{cases}$	
killing1889_EQ0088	022	■ 0.173 ● C|-31	`\left|\begin{array}{cccc} a_{i_{1}} i_{2}} & a_{i_{2}} i_{2}} & \ldots & a_{i_{2}} i_{\rho}} \\ a_{i_{2}} i_{1}} & a_{i_{2}} i_{2}} & \ldots & a_{i_{2}} i_{\rho}} \\ \cdot & \cdot & \cdot & \cdot \\ a_{i_{p}} i_{2}} & a_{i_{p}} i_{2}} & \ldots & a_{i_{p}} i_{p}} \end{array}\right|,`	$\begin{vmatrix} a_{i_1 i_2} & a_{i_2 i_2} & \ldots & a_{i_2 i_\rho} \\ a_{i_2 i_1} & a_{i_2 i_2} & \ldots & a_{i_2 i_\rho} \\ \cdot & \cdot & \cdot & \cdot \\ a_{i_p i_2} & a_{i_p i_2} & \ldots & a_{i_p i_p} \end{vmatrix},$	
killing1889_EQ0166	043	■ 0.180 ● W|-1	`\begin{gathered} \frac{e_{i+x}^{y}}{z} \\ g_{i+x}^{2} \\ z \end{gathered}=\frac{e_{i}^{y}}{g_{i}^{2}}+\frac{e_{x}^{y}}{g_{x}^{2}} \quad \text { u. s. w. }`	$\begin{gathered} \frac{e^y_{i+x}}{z} \\ g^2_{i+x} \\ z \end{gathered} = \frac{e^y_i}{g^2_i} + \frac{e^y_x}{g^2_x} \quad \text{u. s. w.}$	
killing1889_EQ0154 (6)	040	■ 0.185 ● C|-8	`c_{\frac{4+x}{2}} \frac{-4+2}{2} \frac{x+2}{2} c_{\frac{x+2}{2}} \frac{-1-x}{2} \frac{-1+2}{2}=g_{\frac{1+x}{2}}^{2} .`	$c_{\frac{4+x}{2}} \frac{-4+2}{2} \frac{x+2}{2} c_{\frac{x+2}{2}} \frac{-1-x}{2} \frac{-1+2}{2} = g^2_{\frac{1+x}{2}}.$	
killing1889_EQ0080 (3)	019	■ 0.211 ● X|+3	`\left|\begin{array}{cccc} b_{11}-s & b_{12} & \ldots & b_{1 l} \\ b_{21} & b_{22}-s & \ldots & b_{2 l} \\ \cdot & \cdot & \cdot & \cdot \\ b_{l 1} & b_{l 2} & \ldots & b_{u}-s \end{array}\right|`	$\begin{vmatrix} b_{11}-s & b_{12} & \ldots & b_{1l} \\ b_{21} & b_{22}-s & \ldots & b_{2l} \\ \cdot & \cdot & \cdot & \cdot \\ b_{l1} & b_{l2} & \ldots & b_u - s \end{vmatrix}$	
killing1889_EQ0100	027	■ 0.247 ● C|-7	`a_{\mu \sigma}=-1, a_{i \mu}=-1, a_{\mu a}=-2 .`	$a_{\mu\sigma} = -1, a_{i\mu} = -1, a_{\mu a} = -2.$	
killing1889_EQ0137 (2')	034	■ 0.259 ● C|+3	`\left.a_{a t} E_{t}=\sum c_{t, \alpha} c_{t^{\prime} \alpha^{2}}-c_{t a *} c_{t^{\prime} \alpha \alpha}\right) .`	$a_{at}E_t = \sum\left(c_{t,\alpha}c_{t'\alpha^2} - c_{ta*}c_{t'\alpha\alpha}\right).$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix's reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● X crop overran its region ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

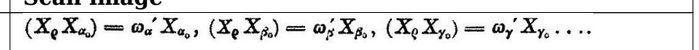
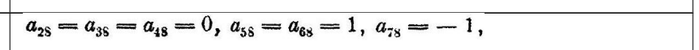
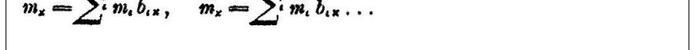
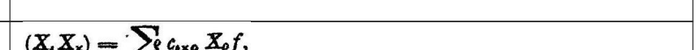
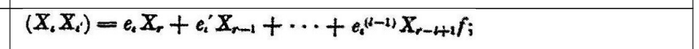
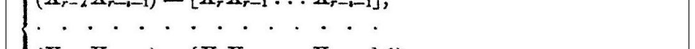
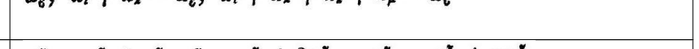
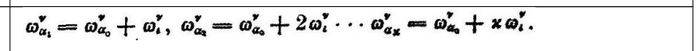
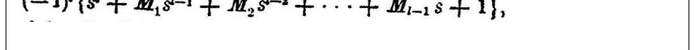
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
killing1889_EQ0103	027	■ 0.266 ● C|-18	`a_{\mu \nu}=-1, a_{\mu, L-1}=a_{L-1, \mu}=-1, a_{\mu l}=a_{l \mu}=-1, a_{l-1, l}=a_{l, L-1}=0 .`	$a_{\mu\nu} = -1, a_{\mu,L-1} = a_{L-1,\mu} = -1, a_{\mu l} = a_{l\mu} = -1, a_{l-1,l} = a_{l,L-1} = 0.$	
killing1889_EQ0113	029	■ 0.273 ● C|-14	`a_{x i} a_{l i}=a_{2 i} a_{l x}, \quad a_{l i-1} a_{x i} =2 a_{l-1, l} a_{l x},`	$a_{xi}a_{li} = a_{2i}a_{lx}, \quad a_{li-1}a_{xi} = 2a_{l-1,l}a_{lx},$	
killing1889_EQ0105	028	■ 0.277 ● C|-3	`s^{2}+s^{n-1}+s^{n-2}+\cdots+s+1,`	$s^2 + s^{n-1} + s^{n-2} + \cdots + s + 1,$	
killing1889_EQ0163	042	■ 0.278 ● C|-3	`X_{r} f, X_{r-1} f \ldots X_{r-i+1} f, X_{\imath} f, X_{-\iota} f, X_{\frac{c+x}{2}} f, \frac{X_{t-x}}{2} f, X_{\frac{-t-x}{2}} f`	$X_r f, X_{r-1} f \ldots X_{r-i+1} f, X_\imath f, X_{-\iota} f, X_{\frac{c+x}{2}} f, \frac{X_{t-x}}{2} f, X_{\frac{-t-x}{2}} f$	
killing1889_EQ0191	047	■ 0.280 ●	`c_{-1}^{c} \frac{1+2+3+4}{2} \frac{-1+2+3+4}{8}`	$c_{-1}^{c} \frac{1+2+3+4}{2} \frac{-1+2+3+4}{8}$	
killing1889_EQ0070	017	■ 0.291 ● C|-11	`a_{a x}=h_{a 1} a_{1 x}+h_{a 2} a_{2 x}+\cdots+h_{a l} a_{i x} \quad \text { för } \quad x=1 \ldots l .`	$a_{ax} = h_{a1}a_{1x} + h_{a2}a_{2x} + \cdots + h_{al}a_{ix} \quad \text{för} \quad x = 1 \ldots l.$	
killing1889_EQ0053 (left1^{1}right)	014	■ 0.299 ● C|-12	`\left(X_{r-1} X_{\alpha}\right)=\omega_{\alpha}{ }^{ \prime} X_{\alpha} f, \ldots\left(X_{r-2+1} X_{\alpha} \right)=\omega_{\alpha}^{(l-1)} X_{\alpha} f .`	$(X_{r-1}X_\alpha) = \omega_\alpha{}' X_\alpha f, \ldots (X_{r-2+1}X_\alpha) = \omega_\alpha^{(l-1)} X_\alpha f.$	
killing1889_EQ0065 (7)	016	■ 0.303 ● W|-2	`\sum_{0}^{l-1} e_{t}^{(\eta)}\left(2 \omega_{\alpha}^{( \nu)}+a_{\alpha i} \omega_{t}^{(\nu)}\right)=0`	$\sum_{0}^{l-1} e_t^{(\eta)} \left(2\omega_\alpha^{(\nu)} + a_{\alpha i}\omega_t^{(\nu)}\right) = 0$	
killing1889_EQ0081 (4)	020	■ 0.307 ● C|+11	`\left|\begin{array}{ccccc} -s-1 & a_{21} & a_{31} & \ldots & a_{l 1} \\ a_{12} s & -s-1 & a_{32} & \ldots & a_{l 2} \\ a_{13} s & a_{23} s & -s-1 & \ldots & a_{23} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{11} s & a_{2 l} s & a_{3 l} s & \ldots & -s-1 \end{array}\right| .`	$\begin{vmatrix} -s-1 & a_{21} & a_{31} & \ldots & a_{l1} \\ a_{12}s & -s-1 & a_{32} & \ldots & a_{l2} \\ a_{13}s & a_{23}s & -s-1 & \ldots & a_{23} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{11}s & a_{2l}s & a_{3l}s & \ldots & -s-1 \end{vmatrix}.$	
killing1889_EQ0061	015	■ 0.312 ● C|-14	`c_{i a 1} c_{i 1 a}, c_{i a a} c_{i 2 a}, \ldots c_{i a 1 } c_{i a}, c_{i a 2} c_{i 2 a} \ldots`	$c_{ia1}c_{i1a}, c_{iaa}c_{i2a}, \ldots c_{ia1}c_{ia}, c_{ia2}c_{i2a} \ldots$	
killing1889_EQ0064	015	■ 0.316 ● C|-15	`\left(X_{i} X_{i^{\prime}}\right)=e_{i} X_{r} f+e_{i}^{ \prime} X_{r-1} f+\cdots+e_{i}^{(i-1)} X_{r-i+1} f,`	$(X_i X_{i'}) = e_i X_r f + e_i' X_{r-1} f + \cdots + e_i^{(i-1)} X_{r-i+1} f,$	
killing1889_EQ0132	032	■ 0.329 ● C|-11	`a_{23}=a_{38}=0, a_{48}=-1, a_{58}=a_{68}=0, a_{78}=-1,`	$a_{23} = a_{38} = 0, a_{48} = -1, a_{58} = a_{68} = 0, a_{78} = -1,$	
killing1889_EQ0159 (11)	041	■ 0.342 ● W|+1	`c_{\frac{t+x}{2}-\frac{t+2}{2} \frac{x+2}{2}}=\sqrt{\frac {p_{x 2} p_{2 x} p_{t 2}}{p_{t x} p_{x i} p_{2 i}}} \frac {\frac{g_{t+x}}{2} \frac{g_{t-2}}{2}}{\frac{g_{x+2}}{2}} .`	$c_{\frac{t+x}{2}-\frac{t+2}{2}\frac{x+2}{2}} = \sqrt{\frac{p_{x2}p_{2x}p_{t2}}{p_{tx}p_{xi}p_{2i}}} \frac{\frac{g_{t+x}}{2}\frac{g_{t-2}}{2}}{\frac{g_{x+2}}{2}}.$	
killing1889_EQ0003	002	■ 0.344 ● C|-10	`\left(\sum \eta_{i} X_{i}, \sum \xi_{i} X_{i}\right)=\sum _{i \times \rho} \eta_{i} \xi_{x} c_{i \times \rho} X_{\rho} f=\omega \sum \xi_{\rho} X_{\rho} f .`	$\left(\sum \eta_i X_i, \sum \xi_i X_i\right) = \sum_{i\times\rho} \eta_i \xi_x c_{i\times\rho} X_\rho f = \omega \sum \xi_\rho X_\rho f.$	
killing1889_EQ0030	009	■ 0.370 ● C|-14	`\sum c_{r-k+2, \alpha, \beta} \omega_{\beta} X_{\beta}- \omega_{\alpha} \sum c_{\gamma-k+2, \alpha, \beta} X_{ \beta}-a_{k+2} \omega_{\alpha}^{(k-1)} X_{\alpha}=0,`	$\sum c_{r-k+2,\alpha,\beta}\omega_\beta X_\beta - \omega_\alpha \sum c_{\gamma-k+2,\alpha,\beta} X_\beta - a_{k+2}\omega_\alpha^{(k-1)} X_\alpha = 0,$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix's reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● X crop overran its region ● not measured

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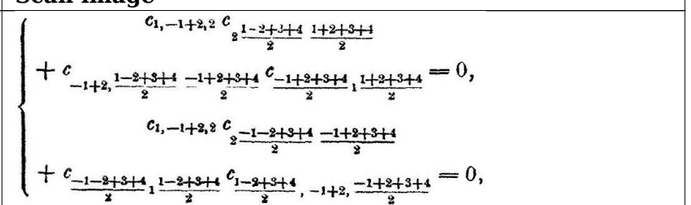
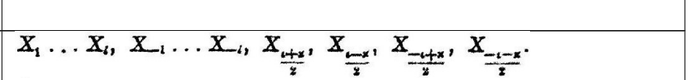
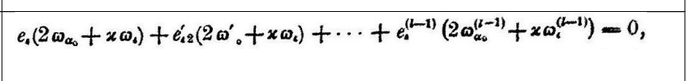
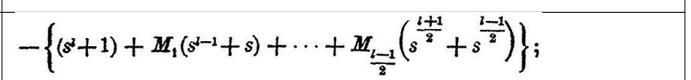
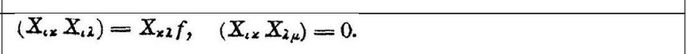
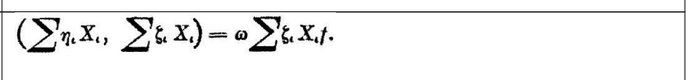
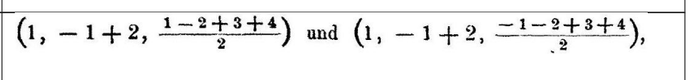
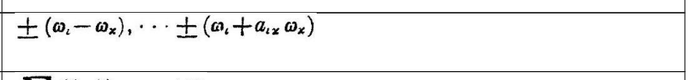
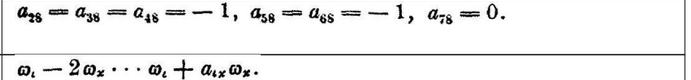
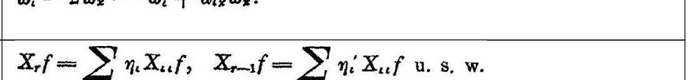
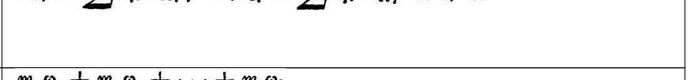
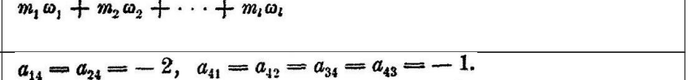
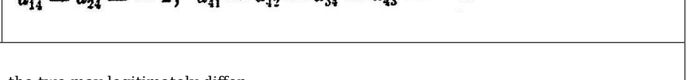
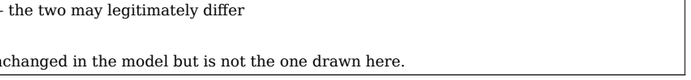
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
killing1889_EQ0036	010	■ 0.379 ● C|-7	`\left(X_{\boldsymbol{\rho}} X_{\alpha_{0}}\right)=\omega_{\alpha}^{\prime} X_{\alpha_{0}},\left(X_{\boldsymbol{\rho}} X_{\beta_{0}}\right)=\omega_{p}^{\prime} X_{\beta_{0}},\left(X_{\boldsymbol{\rho}} X_{\gamma_{0}}\right)=\omega_{\gamma}^{\prime} X_{\gamma_{0}} \ldots .`	$(X_{\boldsymbol{\rho}} X_{\alpha_0}) = \omega'_{\alpha} X_{\alpha_0}, (X_{\boldsymbol{\rho}} X_{\beta_0}) = \omega'_{p} X_{\beta_0}, (X_{\boldsymbol{\rho}} X_{\gamma_0}) = \omega'_{\gamma} X_{\gamma_0} \ldots .$	
killing1889_EQ0129	032	■ 0.407 ● C|-4	`a_{2 \mathrm{~S}}=a_{38}=a_{48}=0, a_{58}=a_{68}=1, a_{78}=-1,`	$a_{2\,\mathrm{S}} = a_{38} = a_{48} = 0, a_{58} = a_{68} = 1, a_{78} = -1,$	
killing1889_EQ0079	019	■ 0.411 ● C|-3	`\bar{m}_{x}=\sum m_{i} b_{i x}, \quad \overline{m_{x}}=\sum \bar{m}_{i} b_{i x} \ldots`	$\bar{m}_x = \sum m_i b_{ix}, \quad \overline{m_x} = \sum \bar{m}_i b_{ix} \ldots$	
killing1889_EQ0092	024	■ 0.442 ● C|-9	`a_{12}^{\prime}=a_{21}^{\prime}=a_{23}^{\prime}=a_{32}^{\prime}=-1, a_{13}^{\prime}=a_{31}^{\prime}=0 ;`	$a'_{12} = a'_{21} = a'_{23} = a'_{32} = -1, a'_{13} = a'_{31} = 0;$	
killing1889_ EQ0067 (9)	016	■ 0.444 ● N|+0	`\sum e_{l}^{(\nu)} \omega_{\alpha}^{(\nu)}=a_{a_{l}} E_{l} .`	$\sum e_l^{(\nu)} \omega_\alpha^{(\nu)} = a_{a_l} E_l.$	
killing1889_EQ0001	002	■ 0.514 ● C|-6	`\left(X_{i} X_{x}\right)=\sum ? c_{t \times p} X_{p} f,`	$(X_i X_x) = \sum ? c_{t\times p} X_p f,$	
killing1889_ EQ0058 (4)	015	■ 0.526 ●	`\left(X_{i} X_{i}^{\prime}\right)=e_{i} X_{r}+e_{i}^{\prime} X_{r-1}+\cdots+e_{i}^{(l-1)} X_{r-1+1} f ;`	$(X_i X'_i) = e_i X_r + e'_i X_{r-1} + \cdots + e_i^{(l-1)} X_{r-1+1} f;$	
killing1889_ EQ0021 (5)	006	■ 0.539 ● W|+0	`\left\{\begin{array}{l} \left(X_{r-2} X_{r-i-1}\right)=\left[X_{r} X_{r-1} \ldots X_{r-i-1}\right], \\ \left.\left(X_{r-2} X_{r-k+1}\right)=\left[X_{r} X_{r-1} \ldots X_{r-k+1}\right] .^{*}\right) \end{array}\right.`	$\begin{cases} (X_{r-2} X_{r-i-1}) = [X_r X_{r-1} \ldots X_{r-i-1}], \\ (X_{r-2} X_{r-k+1}) = [X_r X_{r-1} \ldots X_{r-k+1}] .^{*}) \end{cases}$	
killing1889_EQ0192	047	■ 0.565 ● C|-3	`\omega_{6}, \pi_{i}+\pi_{x}-\omega_{i}, \pi_{i}+\pi_{x}+\pi_{i}+\pi_{\mu}-\omega_{G}`	$\omega_6, \pi_i + \pi_x - \omega_i, \pi_i + \pi_x + \pi_i + \pi_\mu - \omega_G$	
killing1889_EQ0062	015	■ 0.575 ● C|-7	`\omega_{\alpha_{1}}^{\nu}=\omega_{\alpha_{0}}^{\nu}+\omega_{t}^{\nu}, \omega_{\alpha_{2}}^{\nu}=\omega_{\alpha_{0}}^{\nu}+2 \omega_{t}^{\nu} \cdots \omega_{\alpha_{x}}^{\nu}=\omega_{\alpha_{0}}^{\nu}+x \omega_{i}^{\nu} .`	$\omega^{\nu}_{\alpha_1} = \omega^{\nu}_{\alpha_0} + \omega^{\nu}_t, \omega^{\nu}_{\alpha_2} = \omega^{\nu}_{\alpha_0} + 2\omega^{\nu}_t \cdots \omega^{\nu}_{\alpha_x} = \omega^{\nu}_{\alpha_0} + x\omega^{\nu}_i.$	
killing1889_EQ0138	034	■ 0.595 ● S|-1	`=\sum_{v}^{h-1} e_{t}^{(v)} X_{r-v} f .`	$= \sum_{v}^{h-1} e_t^{(v)} X_{r-v} f.$	
killing1889_ EQ0097 (1)	026	■ 0.609 ● N|-2	`a_{l x} a_{l l}=a_{x l} a_{l},`	$a_{lx} a_{ll} = a_{xl} a_{l},$	
killing1889_ EQ0082 (5)	020	■ 0.618 ● W|-1	`(-1)^{t}\left\{s^{t}+M_{1} s^{s^{-1}}+M_{2} s^{l-2}+\cdots+M_{l-1} s+1\right\},`	$(-1)^t \left\{ s^t + M_1 s^{s^{-1}} + M_2 s^{l-2} + \cdots + M_{l-1} s + 1 \right\},$	
killing1889_EQ0031	009	■ 0.631 ● C|-11	`a_{k-2} \omega_{\alpha}^{(k-1)}=0 \text { und } c_{r-k+2, \alpha, \beta}=0 \text { für } \alpha \geqslant \beta .`	$a_{k-2}\omega_\alpha^{(k-1)} = 0$ und $c_{r-k+2,\alpha,\beta} = 0$ für $\alpha \geqslant \beta.$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix's reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

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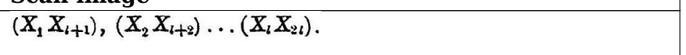
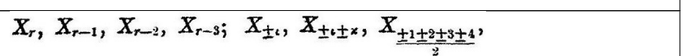
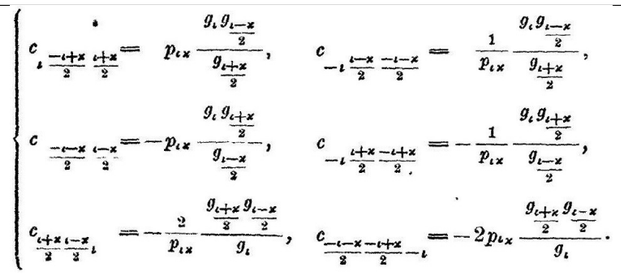
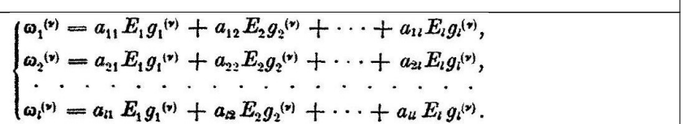
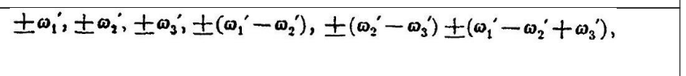
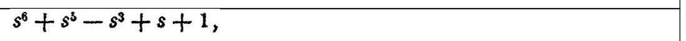
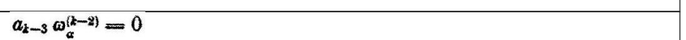
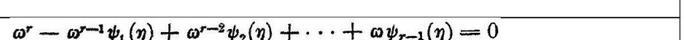
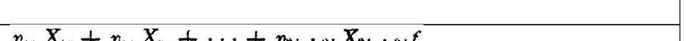
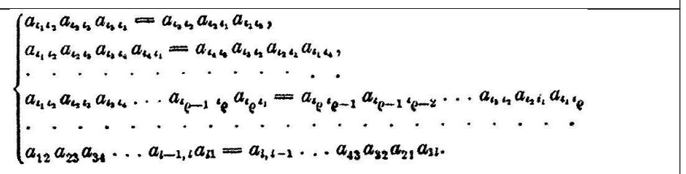
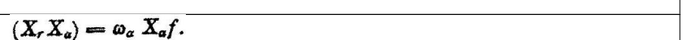
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
killing1889_ EQ0185 (2)	046	■ 0.646 ● C|-35	`\left\{\begin{array}{c} c_{1,-1+2,2} c_{2} \frac{1-2+3+4}{2} \frac{1+2+3+4}{2} \\ +c_{-1+2, \frac{1-2+3+4}{2} -1+2+3+4} c_{-1+2+3+4}{2} \frac{1+2+3+4}{2}=0, \\ c_{1,-1+2,2} c_{2} \frac{-1-2+3+4}{2} \frac{-1+2+3+4}{2} \\ +c_{\frac{-1-2+3+4}{2}} \frac{1-2+3+4}{2} \frac{c_{1-2+3+4}}{2}, -1+2, \frac{-1+2+3+4}{2}=0, \end{array}\right.`	$\begin{cases} c_{1,-1+2,2}c_2\frac{1-2+3+4}{2}\frac{1+2+3+4}{2} \\ +c^2_{-1+2,\frac{1-2+3+4}{2}-1+2+3+4}\frac{c_{-1+2+3+4}}{2}\frac{1+2+3+4}{2}=0, \\ c_{1,-1+2,2}c_2\frac{-1-2+3+4}{2}\frac{-1+2+3+4}{2} \\ +c_{\frac{-1-2+3+4}{2}}\frac{1-2+3+4}{2}\frac{c_{1-2+3+4}}{2},-1+2,\frac{-1+2+3+4}{2}=0, \end{cases}$	
killing1889_EQ0147	039	■ 0.648 ● C|-15	`X_{1} \ldots X_{i}, X_{1} \ldots X_{i}, X_{\frac{i+x}{2}}, X_{\frac{i-x}{2}}, X_{\frac{-i+x}{2}}, X_{\frac{-i-x}{2}} .`	$X_1 \dots X_i, X_1 \dots X_i, X_{\frac{i+x}{2}}, X_{\frac{i-x}{2}}, X_{\frac{-i+x}{2}}, X_{\frac{-i-x}{2}}.$	
killing1889_EQ0063	015	■ 0.659 ● C|-12	`e_{i}\left(2 \omega_{\alpha_{0}}+x \omega_{i}\right)+e_{i2}^{\prime}\left(2 \omega_{0}^{\prime}+x \omega_{i}\right)+\cdots+e_{i}^{(l-1)}\left(2 \omega_{\alpha_{0}}^{(l-1)}+x \omega_{i}^{(l-1)}\right)=0,`	$e_i\left(2\omega_{\alpha_0}+x\omega_i\right)+e'_{i2}\left(2\omega'_0+x\omega_i\right)+\cdots+e_i^{(l-1)}\left(2\omega_{\alpha_0}^{(l-1)}+x\omega_i^{(l-1)}\right)=0,$	
killing1889_ EQ0023 (6)	007	■ 0.674 ● C|-9	`\left(X_{r-i-v} X_{r-i-p}\right)=\left[X_{r} \ldots X_{r-k+1}\right] .`	$(X_{r-i-v}X_{r-i-p}) = [X_r \dots X_{r-k+1}].$	
killing1889_EQ0087	022	■ 0.678 ● W|-1	`-\left\{\left(s^{l}+1\right)+M_{1}\left(s^{l-1}+s\right)+\cdots+M_{\frac{l-1}{2}}\left(s^{\frac{l+1}{2}}+s^{\frac{l-1}{2}}\right)\right\}`	$-\left\{\left(s^l+1\right)+M_1\left(s^{l-1}+s\right)+\cdots+M_{\frac{l-1}{2}}\left(s^{\frac{l+1}{2}}+s^{\frac{l-1}{2}}\right)\right\}$	
killing1889_EQ0162	042	■ 0.689 ● C|-6	`\left(X_{i x} X_{i a}\right)=X_{x a} f, \quad\left(X_{i x} X_{i_{\mu} \mu}\right)=0 .`	$(X_{ix}X_{ia}) = X_{xa}f, \quad \left(X_{ix}X_{i_\mu\mu}\right) = 0.$	
killing1889_ EQ0059 (5)	015	■ 0.705 ● C|-4	`e_{i}^{(\nu)}+e_{i}^{(\nu)}=0 .`	$e_i^{(\nu)} + e_i^{(\nu)} = 0.$	
killing1889_ EQ0037 (1)	011	■ 0.707 ●	`\left(\sum \eta_{t} X_{t}, \sum \xi_{t} X_{t}\right)=\omega \sum \xi_{t} X_{t} t .`	$\left(\sum \eta_t X_t, \sum \xi_t X_t\right) = \omega \sum \xi_t X_t t.$	
killing1889_EQ0184	046	■ 0.718 ● X|+5	`\left(1,-1+2, \frac{1-2+3+4}{2}\right) \text { and }\left(1,-1+2, \frac{-1-2+3+4}{2}\right),`	$\left(1, -1+2, \frac{1-2+3+4}{2}\right) \text{ and } \left(1, -1+2, \frac{-1-2+3+4}{2}\right),$	
killing1889_EQ0139	034	■ 0.734 ● X|+1	`\pm\left(\omega_{\imath}-\omega_{x}\right), \cdots \pm\left(\omega_{\imath}+a_{i x} \omega_{x}\right)`	$\pm(\omega_\imath - \omega_x), \cdots \pm(\omega_\imath + a_{\imath x}\omega_x)$	
killing1889_ EQ0136 (3)	033	■ 0.747 ● C|-5	`\sum e_{i}^{(\eta)} \omega_{i}^{(\eta)}=-2 E_{i}`	$\sum e_i^{(\eta)}\omega_i^{(\eta)} = -2E_i$	
killing1889_EQ0133	032	■ 0.754 ● C|-6	`a_{28}=a_{38}=a_{48}=-1, a_{58}=a_{68}=-1, a_{78}=0 .`	$a_{28} = a_{38} = a_{48} = -1, a_{58} = a_{68} = -1, a_{78} = 0.$	
killing1889_EQ0140	034	■ 0.757 ● S|+0	`\omega_{t}-2 \omega_{x} \cdots \omega_{t}+a_{t x} \omega_{x} .`	$\omega_t - 2\omega_x \cdots \omega_t + a_{tx}\omega_x.$	
killing1889_EQ0142	038	■ 0.777 ● W|-1	`X_{r} f=\sum \eta_{\imath} X_{\imath \imath} f, \quad X_{r-1} f=\sum \eta_{\imath}^{\prime} X_{\imath \imath} f \text { u. s. w. }`	$X_r f = \sum \eta_\imath X_{\imath\imath} f, \quad X_{r-1} f = \sum \eta'_\imath X_{\imath\imath} f \text{ u. s. w.}$	
killing1889_EQ0073	018	■ 0.781 ● S|-2	`m_{1} \omega_{1}+m_{2} \omega_{2}+\cdots+m_{l} \omega_{l}`	$m_1\omega_1 + m_2\omega_2 + \cdots + m_l\omega_l$	
killing1889_EQ0115	030	■ 0.803 ● C|-5	`a_{14}=a_{24}=-2, a_{41}=a_{42}=a_{34}=a_{43}=-1 .`	$a_{14} = a_{24} = -2, a_{41} = a_{42} = a_{34} = a_{43} = -1.$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix's reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● X crop overran its region ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

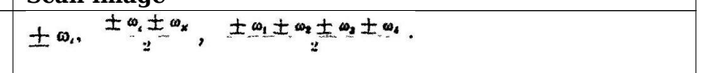
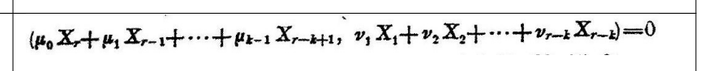
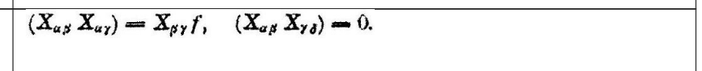
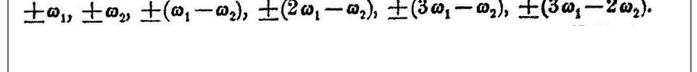
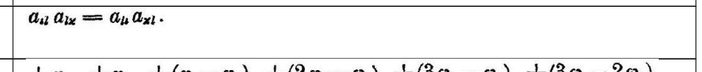
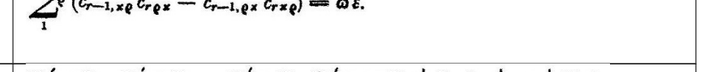
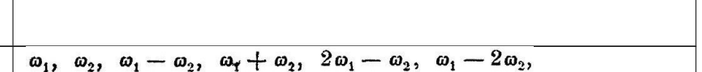
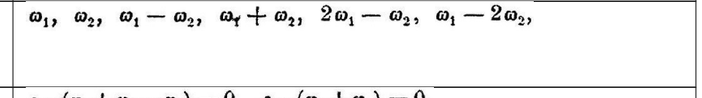
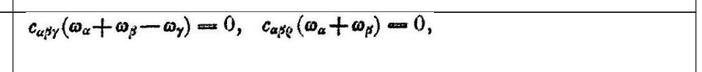
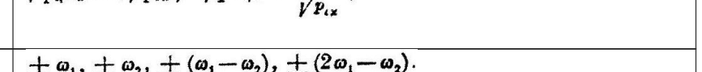
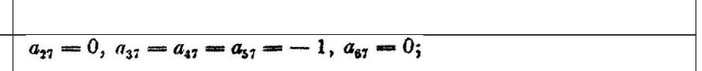
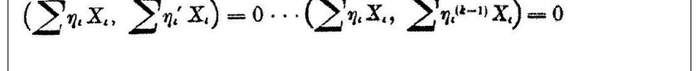
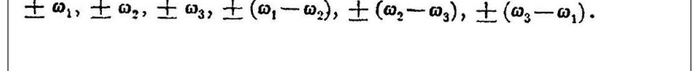
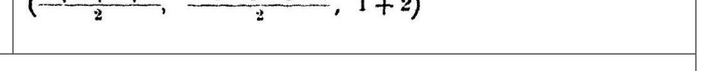
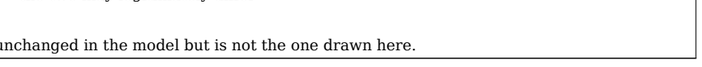
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
killing1889_EQ0056	014	■ 0.851 ● C|-6	`\left(X_{1} X_{i+1}\right),\left(X_{2} X_{i+2}\right) \ldots\left(X_{l} X_{2 l}\right) .`	$\left(X_{1} X_{i+1}\right),\left(X_{2} X_{i+2}\right) \ldots\left(X_{l} X_{2 l}\right) .$	
killing1889_EQ0181	046	■ 0.857 ● X|+6	`X_{r}, X_{r-1}, X_{r-2}, X_{r-3} ; X_{ \pm i}, X_{ \pm \pm x}, X_{ \pm \frac{1 \pm 2 \pm 3 \pm 4}{2}},`	$X_{r}, X_{r-1}, X_{r-2}, X_{r-3} ; X_{\pm i}, X_{\pm \pm x}, X_{\pm \frac{1 \pm 2 \pm 3 \pm 4}{2}},$	
killing1889_EQ0153 (5)	040	■ 0.866 ● C|+213	—	—	
killing1889_EQ0069 (left10^{circ}right)	016	■ 0.867 ● X|+9	`\left\{\begin{array}{l} \omega_{1}^{(\nu)}=a_{11} E_{1} g_{1}^{(\nu)}+a_{12} E_{2} g_{2}^{(\nu)}+\cdots+a_{11} E_{l} g_{l}^{(\nu)}, \\ \omega_{2}^{(\nu)}=a_{21} E_{1} g_{1}^{(\nu)}+a_{22} E_{2} g_{2}^{(\nu)}+\cdots+a_{21} E_{l} g_{l}^{(\nu)}, \\ \omega_{l}^{(\nu)}=a_{11} E_{1} g_{1}^{(\nu)}+a_{22} E_{2} g_{2}^{(\nu)}+\cdots+a_{u} E_{l} g_{l}^{(\nu)} . \end{array}\right.`	$\left\{\begin{array}{l} \omega_{1}^{(\nu)}=a_{11} E_{1} g_{1}^{(\nu)}+a_{12} E_{2} g_{2}^{(\nu)}+\cdots+a_{11} E_{l} g_{l}^{(\nu)}, \\ \omega_{2}^{(\nu)}=a_{21} E_{1} g_{1}^{(\nu)}+a_{22} E_{2} g_{2}^{(\nu)}+\cdots+a_{21} E_{l} g_{l}^{(\nu)}, \\ \omega_{l}^{(\nu)}=a_{11} E_{1} g_{1}^{(\nu)}+a_{22} E_{2} g_{2}^{(\nu)}+\cdots+a_{u} E_{l} g_{l}^{(\nu)}. \end{array}\right.$	
killing1889_EQ0093	024	■ 0.868 ● X|+6	`\pm \omega_{1}^{\prime}, \pm \omega_{2}^{\prime}, \pm \omega_{3}^{\prime}, \pm\left(\omega_{1}^{\prime}-\omega_{2}^{\prime}\right), \pm\left(\omega_{2}^{\prime}-\omega_{3}^{\prime}\right) \pm\left(\omega_{1}^{\prime}-\omega_{2}^{\prime}+\omega_{3}^{\prime}\right),`	$\pm\omega_{1}^{\prime}, \pm\omega_{2}^{\prime}, \pm\omega_{3}^{\prime}, \pm\left(\omega_{1}^{\prime}-\omega_{2}^{\prime}\right), \pm\left(\omega_{2}^{\prime}-\omega_{3}^{\prime}\right) \pm\left(\omega_{1}^{\prime}-\omega_{2}^{\prime}+\omega_{3}^{\prime}\right),$	
killing1889_EQ0110	029	■ 0.876 ● S|-2	`s^{6}+s^{3}-s^{3}+s+1,`	$s^{6}+s^{3}-s^{3}+s+1,$	
killing1889_EQ0032	009	■ 0.879 ● C|-4	`a_{k-3} \omega_{\alpha}^{(k-2)}=0`	$a_{k-3} \omega_{\alpha}^{(k-2)}=0$	
killing1889_EQ0005 (1)	004	■ 0.879 ● C|-5	`\omega^{r}-\omega^{r-1} \psi_{1}(\eta)+\omega^{r-2} \psi_{2}(\eta)+\cdots \pm \omega \psi_{r-1}(\eta)=0`	$\omega^{r}-\omega^{r-1} \psi_{1}(\eta)+\omega^{r-2} \psi_{2}(\eta)+\cdots \pm \omega \psi_{r-1}(\eta)=0$	
killing1889_EQ0146	039	■ 0.880 ● C|-9	`\eta_{12} X_{12}+\eta_{34} X_{34}+\cdots+\eta_{2 i-1,2 i} X_{2 i-1,2 i} f`	$\eta_{12} X_{12}+\eta_{34} X_{34}+\cdots+\eta_{2 i-1,2 i} X_{2 i-1,2 i} f$	
killing1889_EQ0086 (6)	021	■ 0.881 ●	—	—	
killing1889_EQ0052 (1)	013	■ 0.881 ● C|-3	`\left(X_{r} X_{\alpha}\right)=\omega_{\alpha} X_{\alpha} f .`	$\left(X_{r} X_{\alpha}\right)=\omega_{\alpha} X_{\alpha} f.$	
killing1889_EQ0027	008	■ 0.882 ● S|+0	`\left(X_{r} X_{\alpha}\right)=\sum e_{\alpha_{\beta}} X_{ \beta} f`	$\left(X_{r} X_{\alpha}\right)=\sum e_{\alpha_{\beta}} X_{\beta} f$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix's reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● X crop overran its region ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

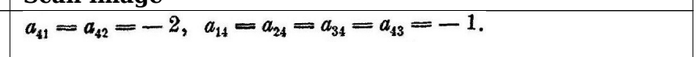
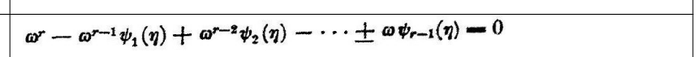
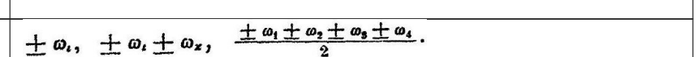
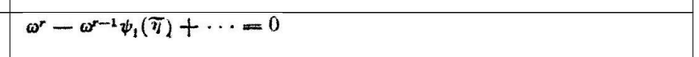
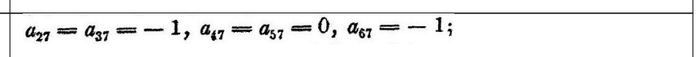
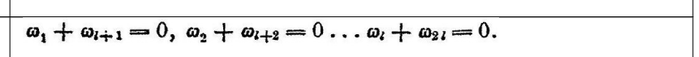
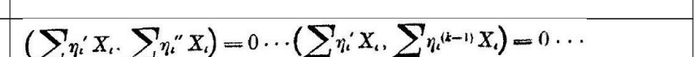
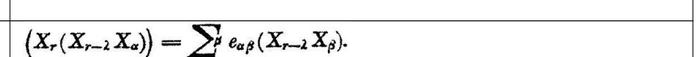
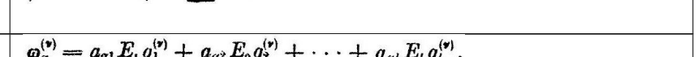
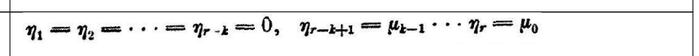
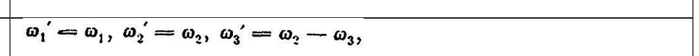
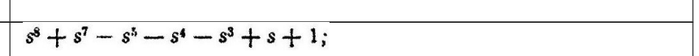
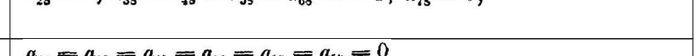
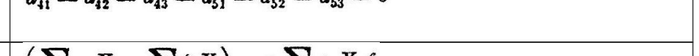
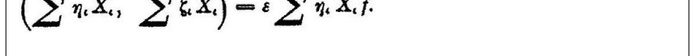
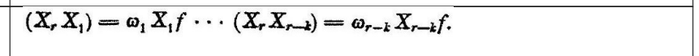
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
killing1889_EQ0189	047	■ 0.883 ● X|+14	`\pm \omega_{1}, \quad \pm \omega_{2} \pm \omega_{x}, \quad \pm \omega_{1} \pm \omega_{2} \pm \omega_{2} \pm \omega_{4} .`	$\pm\omega_1, \quad \pm\omega_2 \pm \omega_x, \quad \pm\omega_1 \pm \omega_2 \pm \omega_2 \pm \omega_4.$	
killing1889_EQ0024	008	■ 0.883 ● C|-6	`\left(\mu_{0} X_{r}+\mu_{1} X_{r-1}+\cdots+\mu_{k-1} X_{r-k+1}, \nu_{1} X_{1}+\nu_{2} X_{2}+\cdots+\nu_{r-k} X_{r-k}\right)=0`	$(\mu_0 X_r + \mu_1 X_{r-1} + \cdots + \mu_{k-1} X_{r-k+1}, \nu_1 X_1 + \nu_2 X_2 + \cdots + \nu_{r-k} X_{r-k}) = 0$	
killing1889_EQ0145	039	■ 0.885 ● C|-4	`\left(X_{\alpha \beta} X_{\alpha \gamma}\right)=X_{\beta \gamma} f, \quad\left(X_{\alpha \beta} X_{\gamma \delta}\right)=0 .`	$(X_{\alpha\beta} X_{\alpha\gamma}) = X_{\beta\gamma} f, \quad (X_{\alpha\beta} X_{\gamma\delta}) = 0.$	
killing1889_EQ0096	025	■ 0.885 ● C|+10	`\pm \omega_{1}, \pm \omega_{2}, \pm\left(\omega_{1}-\omega_{2}\right), \pm\left(2 \omega_{1}-\omega_{2}\right), \pm\left(3 \omega_{1}-\omega_{2}\right), \pm\left(3 \omega_{1}-2 \omega_{2}\right) .`	$\pm\omega_1, \pm\omega_2, \pm(\omega_1-\omega_2), \pm(2\omega_1-\omega_2), \pm(3\omega_1-\omega_2), \pm(3\omega_1-2\omega_2).$	
killing1889_EQ0118	030	■ 0.886 ● C|-3	`a_{l l} a_{l x}=a_{l l} a_{x l} .`	$a_{ll} a_{lx} = a_{ll} a_{xl}.$	
killing1889_EQ0170	044	■ 0.887 ● X|+6	`\pm \omega_{1}, \pm \omega_{2}, \pm\left(\omega_{1}-\omega_{2}\right), \pm\left(2 \omega_{1}-\omega_{2}\right), \pm\left(3 \omega_{1}-\omega_{2}\right), \pm\left(3 \omega_{1}-2 \omega_{2}\right) .`	$\pm\omega_1, \pm\omega_2, \pm(\omega_1-\omega_2), \pm(2\omega_1-\omega_2), \pm(3\omega_1-\omega_2), \pm(3\omega_1-2\omega_2).$	
killing1889_EQ0008	005	■ 0.889 ● C|-6	`\sum_{1}^{m} e\left(c_{r-1, x \rho} c_{r \rho x}-c_{r-1, \rho x} c_{r x \rho}\right)=\omega \varepsilon .`	$\sum_1^m e\left(c_{r-1,x\rho} c_{r\rho x} - c_{r-1,\rho x} c_{rx\rho}\right) = \omega\varepsilon.$	
killing1889_EQ0075	018	■ 0.890 ● C|-8	`m_{2}^{\prime}=m_{2}, m_{3}^{\prime}=m_{3}, \ldots m_{l}^{\prime}=m_{l}, m_{1}^{\prime}=-m_{1}+m_{2} a_{21}+\cdots+m_{l} a_{l 1}`	$m_2' = m_2, m_3' = m_3, \ldots m_l' = m_l, m_1' = -m_1 + m_2 a_{21} + \cdots + m_l a_{l1}$	
killing1889_EQ0171	044	■ 0.891 ● X|+1	`\omega_{1}, \omega_{2}, \omega_{1}-\omega_{2}, \omega_{1}+\omega_{2}, 2 \omega_{1}-\omega_{2}, \omega_{1}-2 \omega_{2},`	$\omega_1, \omega_2, \omega_1-\omega_2, \omega_1+\omega_2, 2\omega_1-\omega_2, \omega_1-2\omega_2,$	
killing1889_EQ0134 (1)	033	■ 0.891 ● C|-4	`c_{\alpha \beta \gamma}\left(\omega_{\alpha}+\omega_{\beta}-\omega_{\gamma}\right)=0, \quad c_{\alpha \beta \varphi}\left(\omega_{\alpha}+\omega_{\beta}\right)=0,`	$c_{\alpha\beta\gamma}(\omega_\alpha+\omega_\beta-\omega_\gamma)=0, \quad c_{\alpha\beta\varphi}(\omega_\alpha+\omega_\beta)=0,$	
killing1889_EQ0013	006	■ 0.891 ● W|-1	`\left(X_{r} X_{\alpha}\right)=\sum e_{\alpha \beta} X_{\beta},`	$(X_r X_\alpha) = \sum e_{\alpha\beta} X_\beta,$	
killing1889_EQ0167	043	■ 0.892 ● X|+1	`V p_{i,-x}=i \sqrt{p_{i x}}, \quad \sqrt{p_{-i, x}}=\frac{-i}{V p_{i x}},`	$Vp_{i,-x} = i\sqrt{p_{ix}}, \quad \sqrt{p_{-i,x}} = \frac{-i}{Vp_{ix}},$	
killing1889_EQ0095	025	■ 0.892 ● C|+4	`\pm \omega_{1}, \pm \omega_{2}, \pm\left(\omega_{1}-\omega_{2}\right), \pm\left(2 \omega_{1}-\omega_{2}\right) .`	$\pm\omega_1, \pm\omega_2, \pm(\omega_1-\omega_2), \pm(2\omega_1-\omega_2).$	
killing1889_EQ0126	031	■ 0.893 ● C|-9	`a_{27}=0, a_{37}=a_{47}=a_{57}=-1, a_{67}=0 ;`	$a_{27}=0, a_{37}=a_{47}=a_{57}=-1, a_{67}=0;$	
killing1889_EQ0038 (2)	011	■ 0.894 ● C|-8	`\left(\sum \eta_{i} X_{i}, \sum \eta_{i}^{\prime} X_{i}\right)=0 \cdots\left(\sum \eta_{i} X_{i}, \sum \eta_{i}^{(k-1)} X_{i}\right)=0`	$\left(\sum \eta_i X_i, \sum \eta_i' X_i\right) = 0 \cdots \left(\sum \eta_i X_i, \sum \eta_i^{(k-1)} X_i\right) = 0$	
killing1889_EQ0090	024	■ 0.895 ●	`\pm \omega_{1}, \pm \omega_{2}, \pm \omega_{3}, \pm\left(\omega_{1}-\omega_{2}\right), \pm\left(\omega_{2}-\omega_{3}\right), \pm\left(\omega_{3}-\omega_{1}\right) .`	$\pm\omega_1, \pm\omega_2, \pm\omega_3, \pm(\omega_1-\omega_2), \pm(\omega_2-\omega_3), \pm(\omega_3-\omega_1).$	
killing1889_EQ0188	047	■ 0.896 ● X|+4	`\left(\frac{1+2+3+4}{2}, \frac{-1-2-3-4}{2}, 1+2\right)`	$\left(\frac{1+2+3+4}{2}, \frac{-1-2-3-4}{2}, 1+2\right)$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix's reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● X crop overran its region ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

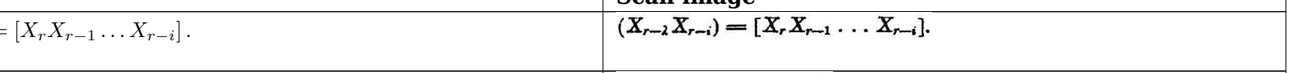
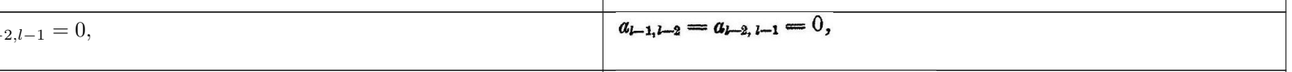
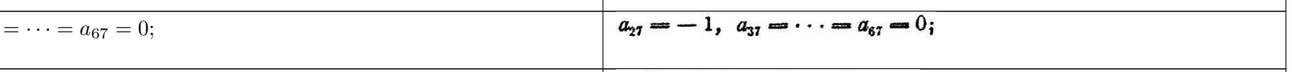
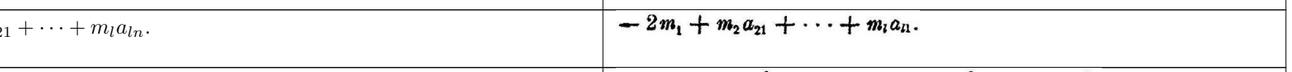
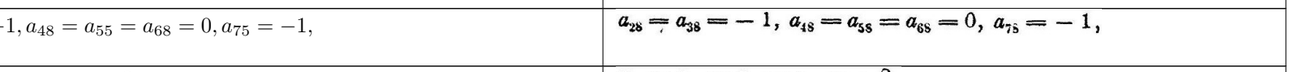
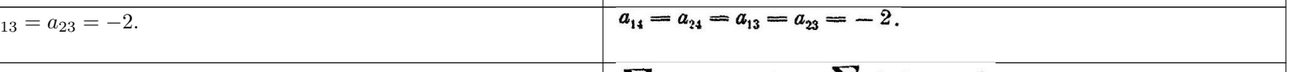
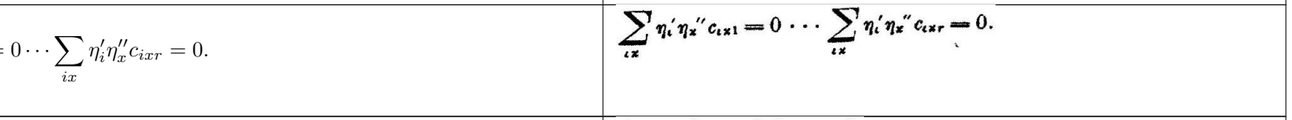
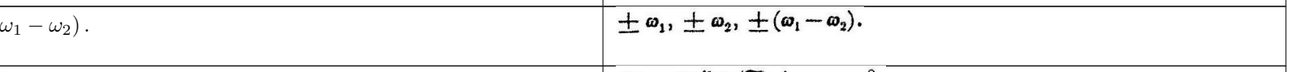
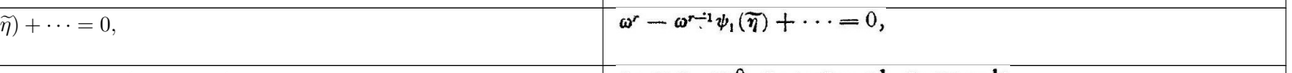
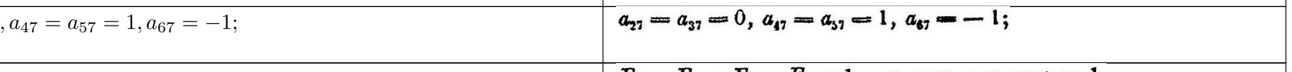
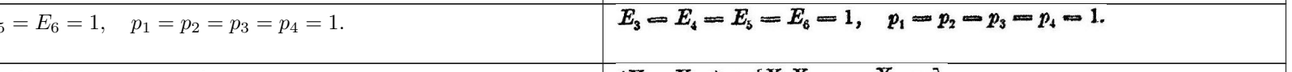
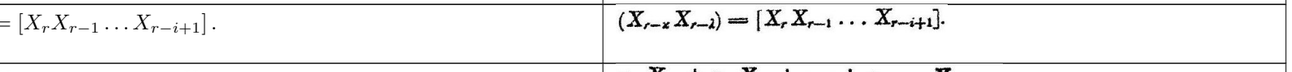
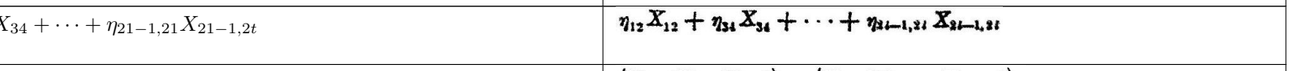
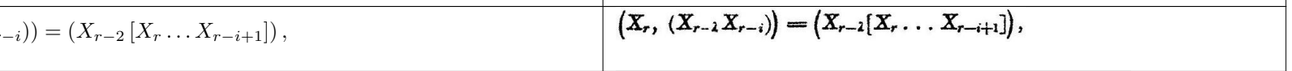
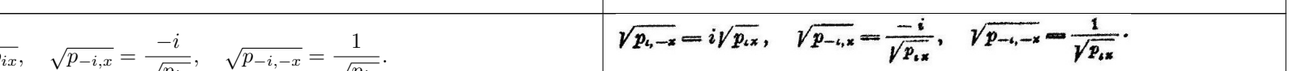
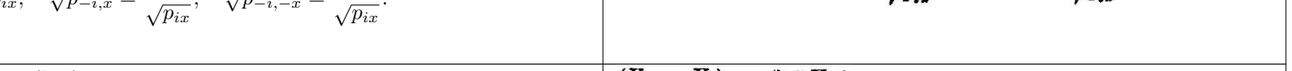
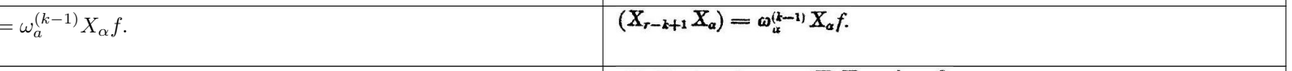
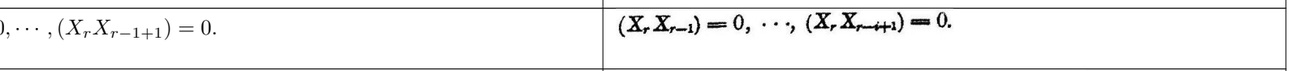
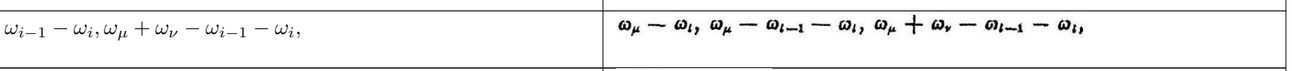
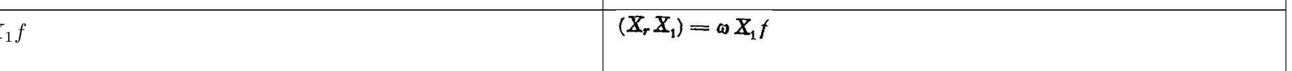
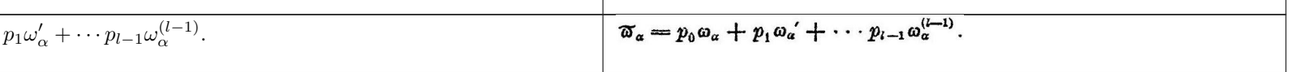
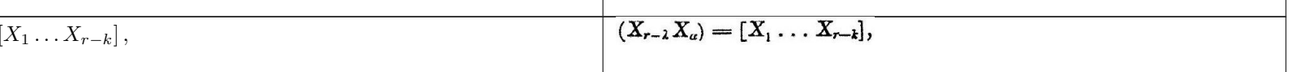
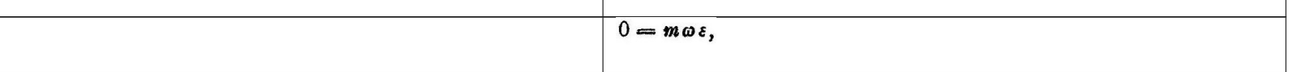
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
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killing1889_EQ0043	011	0.897 C|-4	`\omega^{r}-\omega^{r-1} \psi_{1}(\eta)+\omega^{r-2} \psi_{2}(\eta)-\cdots \pm \omega \psi_{r-1}(\eta)-0`	$\omega^r-\omega^{r-1}\psi_1(\eta)+\omega^{r-2}\psi_2(\eta)-\cdots\pm\omega\psi_{r-1}(\eta)-0$	
killing1889_EQ0176	045	0.897 S|+0	`p_{1} p_{5}=p_{2} p_{4} .`	$p_1p_5=p_2p_4.$	
killing1889_EQ0180	046	0.898 S|-1	`\pm \omega_{i}, \quad \pm \omega_{i} \pm \omega_{x}, \quad \frac{ \pm \omega_{1} \pm \omega_{2} \pm \omega_{s} \pm \omega_{4}}{2} .`	$\pm\omega_i,\quad \pm\omega_i\pm\omega_x,\quad \frac{\pm\omega_1\pm\omega_2\pm\omega_s\pm\omega_4}{2}.$	
killing1889_EQ0083	021	0.899 C|-4	`t_{1}<t_{2}<t_{3}<\cdots<t_{p} ;`	$t_1<t_2<t_3<\cdots<t_p;$	
killing1889_EQ0051	013	0.900 W|-2	`\omega^{r}-\omega^{r-1} \psi_{1}(\widetilde{\eta})+\cdots =0`	$\omega^r-\omega^{r-1}\psi_1(\widetilde{\eta})+\cdots=0$	
killing1889_EQ0127	032	0.901 C|-6	`a_{27}=a_{37}=-1, a_{47}=a_{57}=0, a_{67}=-1 ;`	$a_{27}=a_{37}=-1, a_{47}=a_{57}=0, a_{67}=-1;$	
killing1889_EQ0057	014	0.902 S|-2	`\omega_{1}+\omega_{l+1}=0, \omega_{2}+\omega_{l+2}=0 \ldots \omega_{l}+\omega_{2 l}=0 .`	$\omega_1+\omega_{l+1}=0, \omega_2+\omega_{l+2}=0\ldots\omega_l+\omega_{2l}=0.$	
killing1889_EQ0040 (3)	011	0.902 C|-9	`\left(\sum \eta_{i}^{\prime} X_{i}, \sum \eta_{i}^{\prime \prime} X_{i}\right)=0 \cdots\left(\sum \eta_{i}^{\prime} X_{i}, \sum \eta_{i}^{(k-1)} X_{i}\right)=0 \cdots`	$\left(\sum\eta_i' X_i,\sum\eta_i'' X_i\right)=0\cdots\left(\sum\eta_i' X_i,\sum\eta_i^{(k-1)}X_i\right)=0\cdots$	
killing1889_EQ0006	004	0.903 W|-2	`\left(X_{r} X_{r-1}\right)=\varepsilon X_{r} f,`	$(X_rX_{r-1})=\varepsilon X_r f,$	
killing1889_EQ0016	006	0.903 C|-3	`\left(X_{r}\left(X_{r-2} X_{\alpha}\right)\right)=\sum e_{\alpha \beta}\left(X_{r-2} X_{\beta}\right) .`	$\left(X_r\left(X_{r-2}X_\alpha\right)\right)=\sum e_{\alpha\beta}\left(X_{r-2}X_\beta\right).$	
killing1889_EQ0068 (10)	016	0.904 W|-1	`\omega_{\alpha}^{(\eta)}=a_{\alpha 1} E_{1} g_{1}^{(\eta)}+a_{\alpha 2} E_{2} g_{2}^{(\eta)}+\cdots+a_{a l} E_{l} g_{l}^{(\eta)} .`	$\omega_\alpha^{(\eta)}=a_{\alpha1}E_1g_1^{(\eta)}+a_{\alpha2}E_2g_2^{(\eta)}+\cdots+a_{al}E_lg_l^{(\eta)}.$	
killing1889_EQ0025	008	0.904 C|-9	`\eta_{1}=\eta_{2}=\cdots=\eta_{r-k}=0, \quad \eta_{r-k+1}=\mu_{k-1} \cdots \eta_{r}=\mu_{0}`	$\eta_1=\eta_2=\cdots=\eta_{r-k}=0,\quad \eta_{r-k+1}=\mu_{k-1}\cdots\eta_r=\mu_0$	
killing1889_EQ0091	024	0.905 X|+2	`\omega_{1}^{\prime}=\omega_{1}, \omega_{2}^{\prime}=\omega_{2}, \omega_{3}^{\prime}=\omega_{2}-\omega_{3},`	$\omega_1'=\omega_1,\omega_2'=\omega_2,\omega_3'=\omega_2-\omega_3,$	
killing1889_EQ0112	029	0.906 S|-2	`s^{8}+s^{7}-s^{3}-s^{4}-s^{3}+s+1 ;`	$s^8+s^7-s^3-s^4-s^3+s+1;$	
killing1889_EQ0111	029	0.908 S|-2	`(s+1)\left(s^{6}-s^{3}+1\right),`	$(s+1)\left(s^6-s^3+1\right),$	
killing1889_EQ0014	006	0.909 C|-3	`\left(X_{r}\left(X_{r-x} X_{r-x}\right)\right)=0,`	$\left(X_r\left(X_{r-x}X_{r-x}\right)\right)=0,$	
killing1889_EQ0130	032	0.910 C|-5	`a_{28}=0, a_{38}=a_{48}=a_{58}=a_{68}=-1, a_{78}=0,`	$a_{28}=0, a_{38}=a_{48}=a_{58}=a_{68}=-1, a_{78}=0,$	
killing1889_EQ0089	024	0.910 S|-2	`a_{41}=a_{42}=a_{43}=a_{51}=a_{52}=a_{53}=0`	$a_{41}=a_{42}=a_{43}=a_{51}=a_{52}=a_{53}=0$	
killing1889_EQ0010	005	0.910 C|-8	`\left(\sum \eta_{i} X_{i}, \quad \sum \xi_{i} X_{i}\right)=\varepsilon \sum \eta_{i} X_{i} f .`	$\left(\sum\eta_iX_i,\quad\sum\xi_iX_i\right)=\varepsilon\sum\eta_iX_if.$	
killing1889_EQ0028	008	0.911 C|-5	`\left(X_{r} X_{1}\right)=\omega_{1} X_{1} f \cdots\left(X_{r} X_{r-k}\right)=\omega_{r-k} X_{r-k} f .`	$(X_rX_1)=\omega_1X_1f\cdots(X_rX_{r-k})=\omega_{r-k}X_{r-k}f.$	
killing1889_EQ0019	006	0.911 C|-5	`\left(X_{r},\left(X_{r-2} X_{r-1}\right)\right)=\left[X_{r} \ldots X_{r-i+1}\right]`	$\left(X_r,\left(X_{r-2}X_{r-1}\right)\right)=\left[X_r\ldots X_{r-i+1}\right]$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix's reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • X crop overran its region • not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
killing1889_EQ0020 (4)	006	■ 0.911 ● C|-6	`\left(X_{r-2} X_{r-i}\right)=\left[X_{r} X_{r-1} \ldots X_{r-i}\right] .`	$(X_{r-2}X_{r-i}) = [X_r X_{r-1} \dots X_{r-i}]\,.$	
killing1889_EQ0117	030	■ 0.912 ● C|-7	`a_{l-1, l-2}=a_{l-2, l-1}=0,`	$a_{l-1,l-2} = a_{l-2,l-1} = 0,$	
killing1889_EQ0124	031	■ 0.912 ● C|-8	`a_{27}=-1, a_{37}=\cdots=a_{67}=0 ;`	$a_{27} = -1, a_{37} = \cdots = a_{67} = 0;$	
killing1889_EQ0074	018	■ 0.913 ● C|-3	`-2 m_{1}+m_{2} a_{21}+\cdots+m_{l} a_{l n} .`	$-2m_1 + m_2 a_{21} + \cdots + m_l a_{ln}.$	
killing1889_EQ0131	032	■ 0.913 ● C|-4	`a_{28}=a_{38}=-1, a_{48}=a_{55}=a_{68}=0, a_{75}=-1,`	$a_{28} = a_{38} = -1, a_{48} = a_{55} = a_{68} = 0, a_{75} = -1,$	
killing1889_EQ0120	030	■ 0.914 ● S|-2	`a_{14}=a_{24}=a_{13}=a_{23}=-2 .`	$a_{14} = a_{24} = a_{13} = a_{23} = -2.$	
killing1889_EQ0041 (3′)	011	■ 0.915 ●	`\sum_{i x} \eta_{i}^{\prime} \eta_{x}^{\prime \prime} c_{i x 1}=0 \cdots \sum_{i x} \eta_{i}^{\prime} \eta_{x}^{\prime \prime} c_{i x r}=0 .`	$\sum_{ix} \eta_i' \eta_x'' c_{ix1} = 0 \cdots \sum_{ix} \eta_i' \eta_x'' c_{ixr} = 0.$	
killing1889_EQ0094	025	■ 0.915 ● C|+3	`\pm \omega_{1}, \pm \omega_{2}, \pm\left(\omega_{1}-\omega_{2}\right) .`	$\pm\omega_1, \pm\omega_2, \pm(\omega_1 - \omega_2)\,.$	
killing1889_EQ0047	012	■ 0.916 ● N|+0	`\omega^{r}-\omega^{r-1} \psi_{1}(\widetilde{\eta})+\cdots=0,`	$\omega^r - \omega^{r-1}\psi_1(\widetilde{\eta}) + \cdots = 0,$	
killing1889_EQ0125	031	■ 0.916 ● C|-9	`a_{27}=a_{37}=0, a_{47}=a_{57}=1, a_{67}=-1 ;`	$a_{27} = a_{37} = 0, a_{47} = a_{57} = 1, a_{67} = -1;$	
killing1889_EQ0178	045	■ 0.917 ● C|-9	`E_{3}=E_{4}=E_{5}=E_{6}=1, \quad p_{1}=p_{2}=p_{3}=p_{4}=1 .`	$E_3 = E_4 = E_5 = E_6 = 1, \quad p_1 = p_2 = p_3 = p_4 = 1.$	
killing1889_EQ0015 (2)	006	■ 0.917 ● C|-7	`\left(X_{r-x} X_{r-2}\right)=\left[X_{r} X_{r-1} \ldots X_{r-i+1}\right] .`	$(X_{r-x}X_{r-2}) = [X_r X_{r-1} \dots X_{r-i+1}]\,.$	
killing1889_EQ0144	039	■ 0.918 ● C|-11	`\eta_{12} X_{12}+\eta_{34} X_{34}+\cdots+\eta_{21-1,21} X_{21-1,2 t}`	$\eta_{12}X_{12} + \eta_{34}X_{34} + \cdots + \eta_{21-1,21}X_{21-1,2t}$	
killing1889_EQ0018	006	■ 0.919 ● C|-5	`\left(X_{r},\left(X_{r-2} X_{r-i}\right)\right)=\left(X_{r-2}\left[X_{r} \ldots X_{r-i+1}\right]\right),`	$\left(X_r, (X_{r-2}X_{r-i})\right) = \left(X_{r-2}\left[X_r \dots X_{r-i+1}\right]\right),$	
killing1889_EQ0160 (12)	041	■ 0.920 ● C|-7	`\sqrt{p_{i,-x}}=i \sqrt{p_{i x}}, \quad \sqrt{p_{-i, x}}=\frac{-i}{\sqrt{p_{i x}}}, \quad \sqrt{p_{-i,-x}}=\frac{1}{\sqrt{p_{i x}}} .`	$\sqrt{p_{i,-x}} = i\sqrt{p_{ix}}, \quad \sqrt{p_{-i,x}} = \frac{-i}{\sqrt{p_{ix}}}, \quad \sqrt{p_{-i,-x}} = \frac{1}{\sqrt{p_{ix}}}.$	
killing1889_EQ0029	008	■ 0.921 ● C|-5	`\left(X_{r-k+1} X_{\alpha}\right)=\omega_{a}^{(k-1)} X_{\alpha} f .`	$(X_{r-k+1}X_\alpha) = \omega_a^{(k-1)} X_\alpha f.$	
killing1889_EQ0011	005	■ 0.921 ● C|-9	`\left(X_{r} X_{r-1}\right)=0, \cdots,\left(X_{r} X_{r-1+1}\right)=0 .`	$(X_r X_{r-1}) = 0, \cdots, (X_r X_{r-1+1}) = 0.$	
killing1889_EQ0099	027	■ 0.925 ● C|-4	`\omega_{\mu}-\omega_{i}, \omega_{\mu}-\omega_{i-1}-\omega_{i}, \omega_{\mu}+\omega_{\nu}-\omega_{i-1}-\omega_{i},`	$\omega_\mu - \omega_i, \omega_\mu - \omega_{i-1} - \omega_i, \omega_\mu + \omega_\nu - \omega_{i-1} - \omega_i,$	
killing1889_EQ0007	004	■ 0.929 ● S|-2	`\left(X_{r} X_{1}\right)=\omega X_{1} f`	$(X_r X_1) = \omega X_1 f$	
killing1889_EQ0048	012	■ 0.929 ● W|-2	`\varpi_{\alpha}=p_{0} \omega_{\alpha}+p_{1} \omega_{\alpha}^{\prime}+\cdots p_{l-1} \omega_{\alpha}^{(l-1)} .`	$\varpi_\alpha = p_0\omega_\alpha + p_1\omega_\alpha' + \cdots p_{l-1}\omega_\alpha^{(l-1)}.$	
killing1889_EQ0017 (3)	006	■ 0.930 ● W|-1	`\left(X_{r-2} X_{\alpha}\right)=\left[X_{1} \ldots X_{r-k}\right],`	$(X_{r-2}X_\alpha) = [X_1 \dots X_{r-k}]\,,$	
killing1889_EQ0009	005	■ 0.931 ● W|-1	`0=m \omega \varepsilon,`	$0 = m\omega\varepsilon,$	
killing1889_EQ0128	032	■ 0.932 ● C|-4	`a_{27}=a_{37}=a_{47}=a_{57}=0, a_{67}=-1 .`	$a_{27} = a_{37} = a_{47} = a_{57} = 0, a_{67} = -1.$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix's reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● X crop overran its region ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

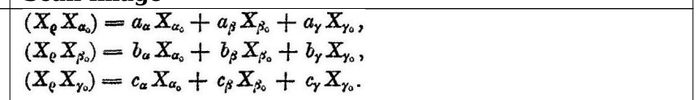
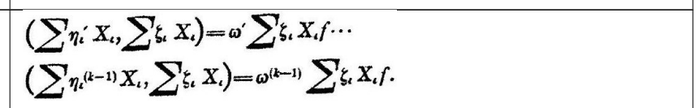
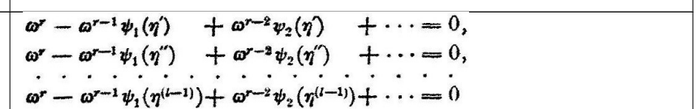
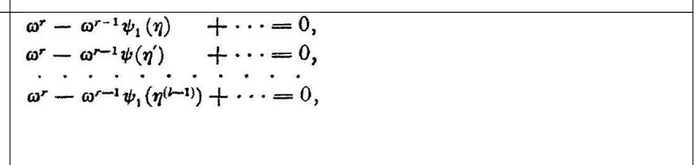
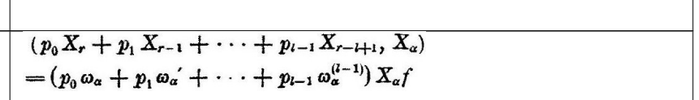
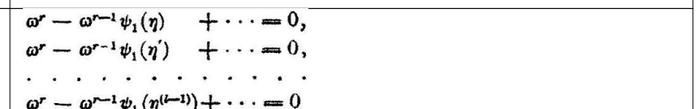
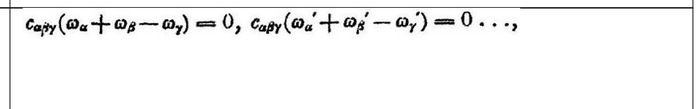
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
killing1889_EQ0004	002	— ● C|+35	\frac{\left.\left\lvert\, \begin{array}{ccc} \sum \eta_{i} c_{i 11}-\omega & \sum \eta_{i} c_{i 21} & \cdots \end{array}\right.\right] \eta_{i} c_{i r 1}}{\sum \eta_{i} c_{i 12}}\left[\left.\begin{array}{ccc} \sum \eta_{i} c_{i 22}-\omega & \cdots & \sum \eta_{i} c_{i r 2} \\ \sum \eta_{i} c_{i 1 r} & \sum \eta_{i} c_{i 2 r} & \cdots \sum \eta_{i} c_{i r r}-\omega \end{array}\right\rvert\, \right.	$\frac{\left\lvert \sum \eta_i c_{i11}-\omega \quad \sum \eta_i c_{i21} \quad \cdots \right] \eta_i c_{ir1}}{\sum \eta_i c_{i12}} \left[\begin{array}{ccc} \sum \eta_i c_{i22}-\omega & \cdots & \sum \eta_i c_{ir2} \\ \sum \eta_i c_{i1r} & \sum \eta_i c_{i2r} & \cdots \sum \eta_i c_{irr}-\omega \end{array} \right\rvert$	$\left\lvert \begin{array}{ccccc} \sum \eta_i c_{i11}-\omega & \sum \eta_i c_{i21} & \cdots & \sum \eta_i c_{ir1} \\ \sum \eta_i c_{i12} & \sum \eta_i c_{i22}-\omega & \cdots & \sum \eta_i c_{ir2} \\ \cdots & \cdots & \cdots & \cdots \\ \sum \eta_i c_{i1r} & \sum \eta_i c_{i2r} & \cdots & \sum \eta_i c_{irr}-\omega \end{array} \right\rvert = \omega^r - \omega^{r-1}\psi_1(\eta) + \omega^{r-2}\psi_2(\eta) - \cdots \pm \omega\,\psi_{r-1}(\eta) = 0$
killing1889_EQ0012	005	— ●	\begin{aligned} & \left(X_{r} X_{r-i}\right)=\left[X_{r} X_{r-1} \cdots X_{r-1+1}\right], \\ & \left(X_{r} X_{r-i-1}\right)=\left[X_{r} X_{r-1} \cdots X_{r-i+1} X_{r-1}\right], \\ & \left(X_{r} X_{r-k+1}\right)=\left[X_{r} X_{r-1} \cdots X_{r-k+2}\right] . \end{aligned}	$\begin{aligned} &(X_r X_{r-i}) = [X_r X_{r-1} \cdots X_{r-1+1}], \\ &(X_r X_{r-i-1}) = [X_r X_{r-1} \cdots X_{r-i+1} X_{r-1}], \\ &(X_r X_{r-k+1}) = [X_r X_{r-1} \cdots X_{r-k+2}]. \end{aligned}$	$\begin{aligned} &(X_r X_{r-i}) = [X_r X_{r-1} \cdots X_{r-i+1}], \\ &(X_r X_{r-i-1}) = [X_r X_{r-1} \cdots X_{r-i+1} X_{r-i}], \\ &\cdots \\ &(X_r X_{r-k+1}) = [X_r X_{r-1} \cdots X_{r-k+2}]. \end{aligned}$
killing1889_EQ0022	007	— ● C|-19	\begin{gathered} \left(X_{r}\left(X_{r-i-v} X_{r-i-p}\right.\right.\right) \\ =\left(X_{r-i-r}\left[X_{r} \ldots X_{r-i-p+i}\right]\right)-\left(X_{r-i-p}\left[X_{r} \ldots X_{r-i-r+1}\right]\right), \end{gathered}	$\begin{gathered} \left(X_r\left(X_{r-i-v} X_{r-i-p}\right.\right.\right) \\ =\left(X_{r-i-r}\left[X_r \ldots X_{r-i-p+i}\right]\right)-\left(X_{r-i-p}\left[X_r \ldots X_{r-i-r+1}\right]\right), \end{gathered}$	$\begin{gathered} \left(X_r\left(X_{r-i-v} X_{r-i-\varrho}\right.\right.\right) \\ =\left(X_{r-i-v}\left[X_r \ldots X_{r-i-\varrho+1}\right]\right)-\left(X_{r-i-\varrho}\left[X_r \ldots X_{r-i-v+1}\right]\right), \end{gathered}$
killing1889_EQ0026	008	— ● C|-15	\begin{gathered} \left(X_{r} X_{r-1}\right)=a_{1} X_{r-2} f, \quad\left(X_{r} X_{r-2}\right)=a_{2} X_{r-3} f \ldots \\ \left(X_{r} X_{r-k+2}\right)=a_{k-2} X_{r-k+1}, \quad \left(X_{r} X_{r-k+1}\right)=0 . \end{gathered}	$\begin{gathered} (X_r X_{r-1}) = a_1 X_{r-2} f, \quad (X_r X_{r-2}) = a_2 X_{r-3} f \ldots \\ (X_r X_{r-k+2}) = a_{k-2} X_{r-k+1}, \quad (X_r X_{r-k+1}) = 0. \end{gathered}$	$\begin{gathered} (X_r X_{r-1}) = a_1 X_{r-2} f, \quad (X_r X_{r-2}) = a_2 X_{r-3} f \ldots \\ (X_r X_{r-k+2}) = a_{k-2} X_{r-k+1}, \quad (X_r X_{r-k+1}) = 0. \end{gathered}$
killing1889_EQ0033	009	— ● C|-9	\begin{aligned} & \left(X_{r} X_{\alpha_{0}}\right)= \omega_{\alpha} X_{\alpha_{0}},\left(X_{r} X_{\alpha_{ 1}}\right)=\omega_{\alpha} X_{\alpha_{1}}+e_{\alpha_{2 } \alpha_{0}} X_{\alpha_{0}}, \ldots \\ & \left(X_{r} X _{\alpha_{x}}\right)=\omega_{\alpha} X_{\alpha_{x}}+e _{\alpha_{x} \alpha_{x-1}} X_{\alpha_{x-1}}+\cdots+e_{ \alpha_{r} \alpha_{0}} X_{\alpha_{0}} . \end{aligned}	$\begin{aligned} &(X_r X_{\alpha_0}) = \omega_\alpha X_{\alpha_0}, (X_r X_{\alpha_1}) = \omega_\alpha X_{\alpha_1} + e_{\alpha_2\alpha_0} X_{\alpha_0}, \ldots \\ &(X_r X_{\alpha_x}) = \omega_\alpha X_{\alpha_x} + e_{\alpha_x\alpha_{x-1}} X_{\alpha_{x-1}} + \cdots + e_{\alpha_r\alpha_0} X_{\alpha_0}. \end{aligned}$	$\begin{aligned} &(X_r X_{\alpha_0}) = \omega_\alpha X_{\alpha_0}, (X_r X_{\alpha_1}) = \omega_\alpha X_{\alpha_1} + e_{\alpha_1\alpha_0} X_{\alpha_0}, \ldots \\ &(X_r X_{\alpha_x}) = \omega_\alpha X_{\alpha_x} + e_{\alpha_x\alpha_{x-1}} X_{\alpha_{x-1}} + \cdots + e_{\alpha_r\alpha_0} X_{\alpha_0}. \end{aligned}$
killing1889_EQ0034	010	— ● C|-15	\begin{aligned} & \left(X_{e} X_{\alpha_{0}}\right)= \omega_{u}^{\prime} X_{\alpha_{0}},\left(X_{i} X_{\alpha_ {1}}\right)=\omega_{u}^{\prime} X_{\alpha_{1}}+e_{\alpha _{1} \alpha_{0}}^{\prime} X_{\alpha_{0}}, \ldots \\ & \left(X_{\ell} X_{\alpha_{x}}\right)=\omega_{u}^{\prime} X_{\alpha_{x}}+e_{\alpha_{x} \alpha_{x-1}}^{\prime} X_{\alpha_{x-1}}+\cdots+e_{\alpha_{x} \alpha_{0}} X_{\alpha_{0}} f . \end{aligned}	$\begin{aligned} &(X_e X_{\alpha_0}) = \omega'_u X_{\alpha_0}, (X_i X_{\alpha_1}) = \omega'_u X_{\alpha_1} + e'_{\alpha_1\alpha_0} X_{\alpha_0}, \ldots \\ &(X_\ell X_{\alpha_x}) = \omega'_u X_{\alpha_x} + e'_{\alpha_x\alpha_{x-1}} X_{\alpha_{x-1}} + \cdots + e_{\alpha_x\alpha_0} X_{\alpha_0} f. \end{aligned}$	$\begin{aligned} &(X_\varrho X_{\alpha_0}) = \omega'_u X_{\alpha_0}, (X_\varrho X_{\alpha_1}) = \omega'_u X_{\alpha_1} + e'_{\alpha_1\alpha_0} X_{\alpha_0}, \ldots \\ &(X_\varrho X_{\alpha_x}) = \omega'_u X_{\alpha_x} + e'_{\alpha_x\alpha_{x-1}} X_{\alpha_{x-1}} + \cdots + e'_{\alpha_x\alpha_0} X_{\alpha_0} f. \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

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Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● X crop overran its region ● not measured

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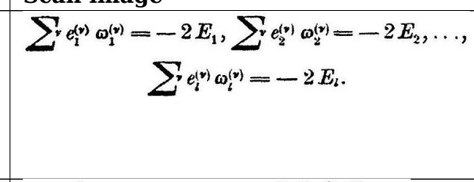
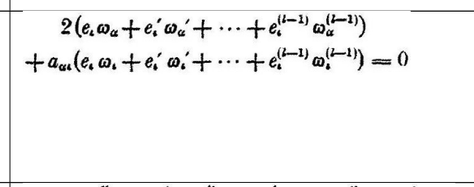
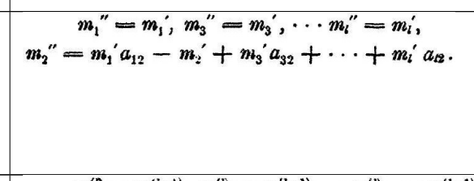
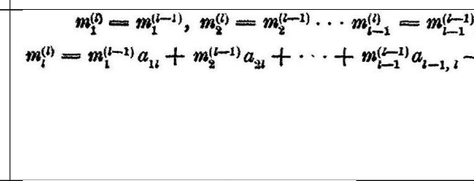
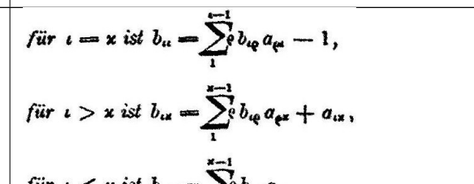
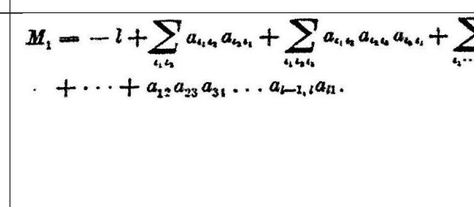
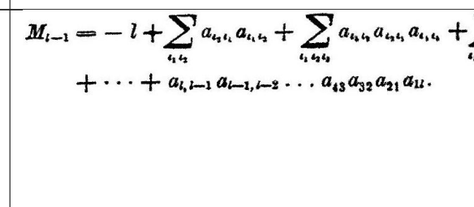
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
killing1889_EQ0035	010	— ● C|-13	\begin{aligned} & \left(X_{\rho} X_{\alpha_{0}}\right)=a_{\alpha} X_{\alpha_{0}}+a_{\beta} X_{\beta_{0}}+a_{\gamma} X_{\gamma_{0}}, \\ & \left(X_{0} X_{\beta_{0}}\right)=b_{a} X_{\alpha_{0}}+b_{\beta} X_{\beta_{0}}+b_{\gamma} X_{\gamma_{0}}, \\ & \left(X_{\rho} X_{\gamma_{0}}\right)=c_{\alpha} X_{\alpha_{0}}+c_{\beta} X_{\beta_{0}}+c_{\gamma} X_{\gamma_{0}} . \end{aligned}	$\begin{aligned} (X_\rho X_{\alpha_0}) &= a_\alpha X_{\alpha_0} + a_\beta X_{\beta_0} + a_\gamma X_{\gamma_0}, \\ (X_0 X_{\beta_0}) &= b_\alpha X_{\alpha_0} + b_\beta X_{\beta_0} + b_\gamma X_{\gamma_0}, \\ (X_\rho X_{\gamma_0}) &= c_\alpha X_{\alpha_0} + c_\beta X_{\beta_0} + c_\gamma X_{\gamma_0}. \end{aligned}$	
killing1889_ EQ0042 (4)	011	— ● C|-17	\begin{aligned} & \left(\sum \eta_{i}^{\prime} X_{i}, \sum \xi_{i} X_{i}\right)=\omega^{\prime} \sum \xi_{i} X_{i} f \cdots \\ & \left(\sum \eta_{i}{ }^{(k-1)} X_{i}, \sum \xi_{i} X_{i}\right)=\omega^{(k-1)} \sum \xi_{i} X_{i} f . \end{aligned}	$\begin{aligned} \left(\sum \eta_i' X_i, \sum \xi_i X_i\right) &= \omega' \sum \xi_i X_i f \cdots \\ \left(\sum \eta_i{}^{(k-1)} X_i, \sum \xi_i X_i\right) &= \omega^{(k-1)} \sum \xi_i X_i f. \end{aligned}$	
killing1889_EQ0044	012	— ●	\begin{aligned} & \omega^{r}-\omega^{r-1} \psi_{1}\left(\eta^{\prime}\right)+\omega^{r-2} \psi_{2}(\eta)+\cdots=0, \\ & \omega^{r}-\omega^{r-1} \psi_{1}\left(\eta^{\prime \prime}\right)+\omega^{r-2} \psi_{2}\left(\eta^{\prime \prime}\right)+\cdots=0, \\ & \omega^{r}-\omega^{r-1} \psi_{1}\left(\eta^{(i-1)}\right)+\omega^{r-2} \psi_{2}\left(\eta^{(i-1)}+\cdots=0 \end{aligned}	$\begin{aligned} \omega^r - \omega^{r-1}\psi_1(\eta') + \omega^{r-2}\psi_2(\eta) + \cdots &= 0, \\ \omega^r - \omega^{r-1}\psi_1(\eta'') + \omega^{r-2}\psi_2(\eta'') + \cdots &= 0, \\ \omega^r - \omega^{r-1}\psi_1\left(\eta^{(i-1)}\right) + \omega^{r-2}\psi_2\left(\eta^{(i-1)}\right. + \cdots &= 0 \end{aligned}$	
killing1889_EQ0045	012	— ● X|+3	\begin{aligned} & \omega^{r}-\omega^{r-1} \psi_{1}(\eta)+\cdots=0, \\ & \omega^{r}-\omega^{r-1} \psi\left(\eta^{\prime}\right)+\cdots=0, \\ & \omega^{r}-\omega^{r-1} \psi_{1}\left(\eta^{(2-1)}\right)+\cdots=0, \\ & \omega^{r}-\omega^{r-1} \psi_{1}\left(\eta^{(2-1)}\right)+\cdots=0, \end{aligned}	$\begin{aligned} \omega^r - \omega^{r-1}\psi_1(\eta) + \cdots &= 0, \\ \omega^r - \omega^{r-1}\psi\left(\eta'\right) + \cdots &= 0, \\ \omega^r - \omega^{r-1}\psi_1\left(\eta^{(2-1)}\right) + \cdots &= 0, \\ \omega^r - \omega^{r-1}\psi_1\left(\eta^{(2-1)}\right) + \cdots &= 0, \end{aligned}$	
killing1889_EQ0049	012	— ● C|-6	\begin{aligned} & \left(p_{0} X_{r}+p_{1} X_{r-1}+\cdots+p_{l-1} X_{r-1+1}, X_{\alpha}\right) \\ & =\left(p_{0} \omega_{\alpha}+p_{1} \omega_{\alpha}^{\prime}+\cdots+p_{l-1} \omega_{\alpha}^{(l-1)}\right) X_{\alpha} f \end{aligned}	$\begin{aligned} &\left(p_0 X_r + p_1 X_{r-1} + \cdots + p_{l-1} X_{r-l+1}, X_\alpha\right) \\ &= \left(p_0 \omega_\alpha + p_1 \omega_\alpha' + \cdots + p_{l-1} \omega_\alpha^{(l-1)}\right) X_\alpha f \end{aligned}$	
killing1889_EQ0050	013	— ● W|+0	\begin{aligned} & \omega^{r}-\omega^{r-1} \psi_{1}(\eta)+\cdots=0, \\ & \omega^{r}-\omega^{r-1} \psi_{1}\left(\eta^{\prime}\right)+\cdots=0, \\ & \omega^{r}-\omega^{r-1} \psi_{1}\left(\eta^{(1-1)}\right)+\cdots=0 \end{aligned}	$\begin{aligned} \omega^r - \omega^{r-1}\psi_1(\eta) + \cdots &= 0, \\ \omega^r - \omega^{r-1}\psi_1\left(\eta'\right) + \cdots &= 0, \\ \omega^r - \omega^{r-1}\psi_1\left(\eta^{(1-1)}\right) + \cdots &= 0 \end{aligned}$	
killing1889_ EQ0054 (2)	014	— ● C|-8	c_{\alpha \beta \gamma}\left(\omega_{\alpha}+\omega_{\beta}-\omega_{\gamma}\right)=0, \quad c_{\alpha \beta \gamma}\left(\omega_{\alpha}^{\prime}+\omega_{\beta}^{\prime}-\omega_{\gamma}^{\prime}\right)=0 \ldots,	$c_{\alpha\beta\gamma}\left(\omega_\alpha + \omega_\beta - \omega_\gamma\right) = 0, \quad c_{\alpha\beta\gamma}\left(\omega_\alpha' + \omega_\beta' - \omega_\gamma'\right) = 0 \ldots,$	
killing1889_ EQ0055 (3)	014	— ● C|-3	c_{\alpha \beta \rho}\left(\omega_{\alpha}+\omega_{\beta}\right)=0, \quad c_{\alpha \beta \rho}\left(\omega_{\alpha}^{\prime}+\omega_{\beta}^{\prime}\right)=0 \ldots .	$c_{\alpha\beta\rho}\left(\omega_\alpha + \omega_\beta\right) = 0, \quad c_{\alpha\beta\rho}\left(\omega_\alpha' + \omega_\beta'\right) = 0 \ldots.$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix's reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● X crop overran its region ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

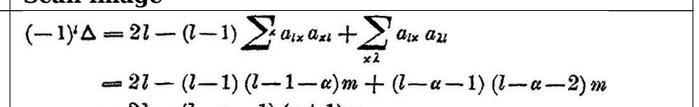
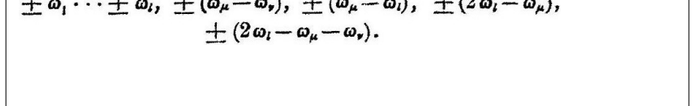
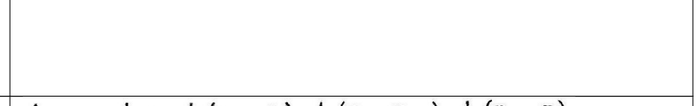
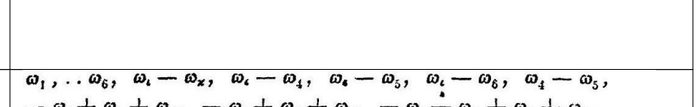
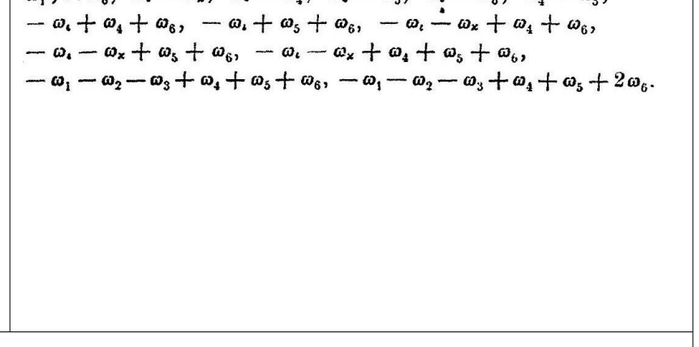
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
killing1889_ EQ00066 (8)	016	— X|+1	\begin{gathered} \sum e_{1}^{(\nu)} \omega_{1}^{(\nu)}=-2 E_{1}, \sum e_{2}^{(\nu)} \omega_{2}^{(\nu)}=-2 E_{2}, \ldots, \\ \sum e_{l}^{(\nu)} \omega_{l}^{(\nu)}=-2 E_{l} . \end{gathered}	$\sum e_1^{(\nu)}\omega_1^{(\nu)} = -2E_1, \sum e_2^{(\nu)}\omega_2^{(\nu)} = -2E_2, \ldots,$ $\sum e_l^{(\nu)}\omega_l^{(\nu)} = -2E_l.$	
killing1889_ EQ00072 (1)	018	— C|-7	\begin{gathered} 2\left(e_{t} \omega_{a}+e_{t}^{\prime} \omega_{c}^{\prime}+\cdots+e_{t}^{(i-1)} \omega_{a}^{(2-1)}\right) \\ +a_{a i}\left(e_{t} \omega_{t}+e_{t}^{\prime} \omega_{t}^{\prime}+\cdots+e_{t}^{(2-1)} \omega_{i}^{(2-1)}\right)=0 \end{gathered}	$2\left(e_t\omega_a + e_t'\omega_c' + \cdots + e_t^{(i-1)}\omega_a^{(2-1)}\right)$ $+a_{ai}\left(e_t\omega_t + e_t'\omega_t' + \cdots + e_t^{(2-1)}\omega_i^{(2-1)}\right) = 0$	
killing1889_EQ0076	018	— C|-7	\begin{gathered} m_{1}^{\prime \prime}=m_{1}^{\prime}, m_{3}^{\prime \prime}=m_{3}^{\prime}, \cdots m_{l}^{\prime \prime}=m_{l}^{\prime}, \\ m_{2}^{\prime \prime}=m_{1}^{\prime} a_{12}-m_{2}^{\prime}+m_{3}^{\prime} a_{32}+\cdots+m_{l}^{\prime} a_{i 2} . \end{gathered}	$m_1'' = m_1', m_3'' = m_3', \cdots m_l'' = m_l',$ $m_2'' = m_1'a_{12} - m_2' + m_3'a_{32} + \cdots + m_l'a_{i2}.$	
killing1889_EQ0077	018	—	\begin{gathered} m_{1}^{(l)}=m_{1}^{(l-1)}, m_{2}^{(l)}=m_{2}^{(l-1)} \cdots m_{l-1}^{(l)}=m_{l-1}^{(l-1)}, \\ m_{l}^{(l)}=m_{1}^{(l-1)} a_{1 l}+m_{2}^{(l-1)} a_{2 l}+\cdots+m_{l-1}^{(l-1)} a_{l-1, l}-m_{l}^{(l-1)} . \end{gathered}	$m_1^{(l)} = m_1^{(l-1)}, m_2^{(l)} = m_2^{(l-1)} \cdots m_{l-1}^{(l)} = m_{l-1}^{(l-1)},$ $m_l^{(l)} = m_1^{(l-1)}a_{1l} + m_2^{(l-1)}a_{2l} + \cdots + m_{l-1}^{(l-1)}a_{l-1,l} - m_l^{(l-1)}.$	
killing1889_ EQ00078 (2)	019	— C|-24	\begin{aligned} & \text { für } \iota=x \text { ist } b_{u}=\sum_{1}^{\infty} b_{b_{e}}^{-1} a_{e x}-1, \\ & \text { für } \iota>x \text { ist } b_{c x}=\sum_{1}^{x-1} b_{c e} a_{e x}+a_{i x}, \\ & \text { für } \iota<x \text { ist } b_{c x}=\sum_{i}^{x-1} b_{c \rho} a_{e x} . \end{aligned}	für $\iota = x$ ist $b_u = \sum_1^{\infty} b_{b_e}^{-1} a_{ex} - 1,$ für $\iota > x$ ist $b_{cx} = \sum_1^{x-1} b_{ce}a_{ex} + a_{ix},$ für $\iota < x$ ist $b_{cx} = \sum_i^{x-1} b_{c\rho}a_{ex}.$	
killing1889_EQ0084	021	— C|-29	\begin{aligned} & M_{1}=-l+\sum_{4 l_{2}} a_{4 l_{4}} a_{4 l_{4}}+\sum_{4 l_{4} l_{4}} a_{l_{4} l_{4}} a_{4 l_{4}} a_{4 l_{4}}+\sum_{l_{4} \cdots l_{4}} a_{4 l_{4}} a_{4 l_{4}} a_{4 l_{4}} a_{4 l_{4}} a_{4} a_{4} \\ & +\cdots+a_{12} a_{23} a_{31} \cdots a_{i-1,1} a_{11} . \end{aligned}	$M_1 = -l + \sum_{4l_2} a_{4l_4}a_{4l_4} + \sum_{4l_4l_4} a_{l_4l_4}a_{4l_4}a_{4l_4} + \sum_{l_4\cdots l_4} a_{4l_4}a_{4l_4}a_{4l_4}a_{4l_4}a_4a_4$ $+ \cdots + a_{12}a_{23}a_{31}\cdots a_{i-1,1}a_{11}.$	
killing1889_EQ0085	021	— C|-26	\begin{aligned} M_{l-1}= & -l+\sum_{l_{1} l_{2}} a_{l_{2} l_{1}} a_{l_{1} l_{2}}+\sum_{l_{2} l_{2} l_{3}} a_{l_{3} l_{4}} a_{l_{2} l_{2}} a_{l_{4} l_{4}}+\sum_{l_{4}+l_{4}} a_{l_{4} l_{4}} a_{l_{4} l_{4}} a_{l_{4} l_{4}} a_{l_{4} l_{4}} \\ & +\cdots+a_{l_{4}-1} a_{l_{1-1}, l_{2}} \cdots a_{l_{4} 3} a_{3_{2}} a_{2_{1}} a_{11} . \end{aligned}	$M_{l-1} = -l + \sum_{l_1l_2} a_{l_2l_1}a_{l_1l_2} + \sum_{l_2l_2l_3} a_{l_3l_4}a_{l_2l_2}a_{l_4l_4} + \sum_{l_4+l_4} a_{l_4l_4}a_{l_4l_4}a_{l_4l_4}a_{l_4l_4}$ $+ \cdots + a_{l_4-1}a_{l_{1-1},l_2}\cdots a_{l_43}a_{3_2}a_{2_1}a_{11}.$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

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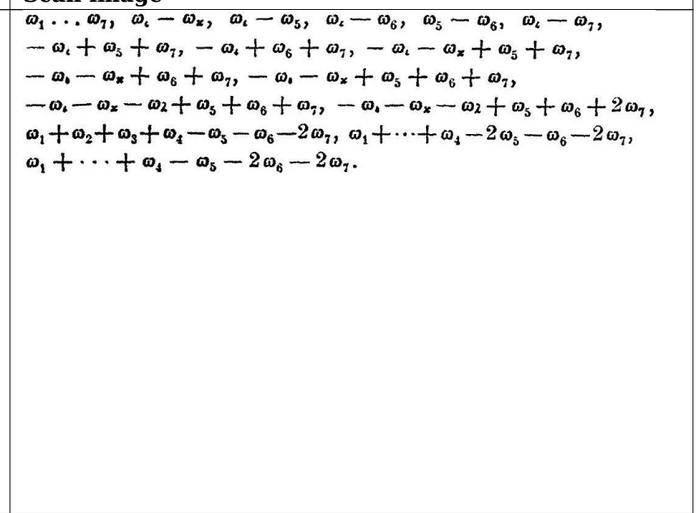
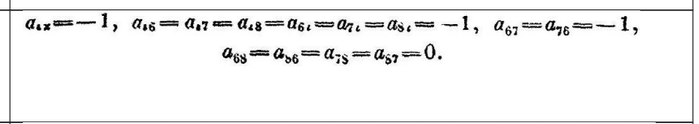
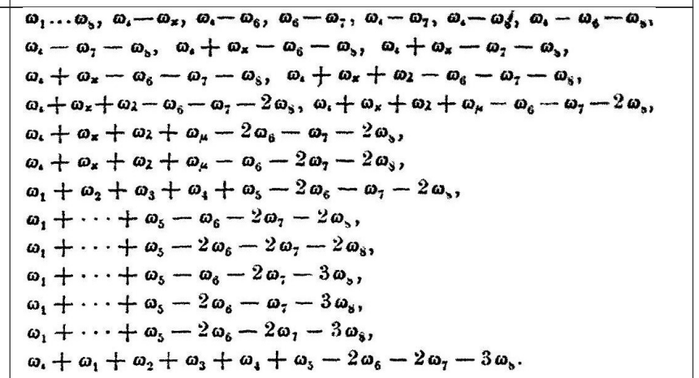
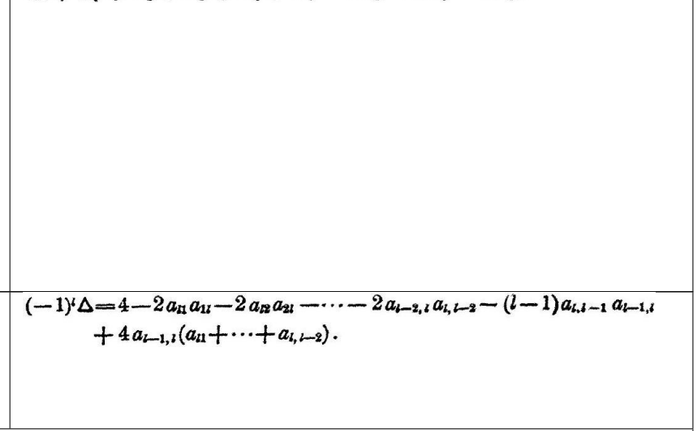
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
killing1889_EQ0098	026	— W|-2	\begin{aligned} (-1)^{l} \Delta &=2 l-(l-1) \sum^{k} a_{l x} a_{x l}+\sum_{x k} a_{l x} a_{2 l} \\ &=2 l-(l-1)(l-1-\alpha) m+(l-\alpha-1)(l-\alpha-2) m \\ &=2 l-(l-\alpha-1)(\alpha+1) m . \end{aligned}	$$\begin{aligned}(-1)^{l}\Delta &= 2l-(l-1)\sum^{k} a_{lx}a_{xl}+\sum_{xk} a_{lx}a_{2l}\\ &= 2l-(l-1)(l-1-\alpha)m+(l-\alpha-1)(l-\alpha-2)m\\ &= 2l-(l-\alpha-1)(\alpha+1)m.\end{aligned}$$	
killing1889_EQ0101	027	— X|+8	\begin{gathered} \pm \omega_{1} \cdots \pm \omega_{l}, \pm\left(\omega_{\mu}-\omega_{\nu}\right), \pm\left(\omega_{\mu}-\omega_{l}\right), \pm\left(2 \omega_{l}-\omega_{\mu}\right), \\ \pm\left(2 \omega_{l}-\omega_{\mu}-\omega_{\nu}\right) . \end{gathered}	$$\begin{gathered}\pm\omega_{1}\cdots\pm\omega_{l},\ \pm(\omega_{\mu}-\omega_{\nu}),\ \pm(\omega_{\mu}-\omega_{l}),\ \pm(2\omega_{l}-\omega_{\mu}),\\ \pm(2\omega_{l}-\omega_{\mu}-\omega_{\nu}).\end{gathered}$$	
killing1889_EQ0102	027	— X|+3	\begin{gathered} \pm \omega_{1} \cdots \pm \omega_{i}, \pm\left(\omega_{\mu}-\omega_{\nu}\right), \pm\left(\omega_{\mu}-\omega_{i}\right), \pm\left(2 \omega_{\mu}-\omega_{i}\right), \\ \pm\left(\omega_{\mu}+\omega_{\nu}-\omega_{i}\right) . \end{gathered}	$$\begin{gathered}\pm\omega_{1}\cdots\pm\omega_{i},\ \pm(\omega_{\mu}-\omega_{\nu}),\ \pm(\omega_{\mu}-\omega_{i}),\ \pm(2\omega_{\mu}-\omega_{i}),\\ \pm(\omega_{\mu}+\omega_{\nu}-\omega_{i}).\end{gathered}$$	
killing1889_EQ0104	028	—	\begin{gathered} \pm \omega_{1} \cdots \pm \omega_{l}, \pm\left(\omega_{\mu}-\omega_{\nu}\right), \pm\left(\omega_{\mu}-\omega_{l-1}\right), \pm\left(\omega_{\mu}-\omega_{l}\right), \\ \pm\left(\omega_{\mu}-\omega_{l-1}-\omega_{l}\right), \pm\left(\omega_{\mu}+\omega_{\nu}-\omega_{l-1}-\omega_{l}\right) . \end{gathered}	$$\begin{gathered}\pm\omega_{1}\cdots\pm\omega_{l},\ \pm(\omega_{\mu}-\omega_{\nu}),\ \pm(\omega_{\mu}-\omega_{l-1}),\ \pm(\omega_{\mu}-\omega_{l}),\\ \pm(\omega_{\mu}-\omega_{l-1}-\omega_{l}),\ \pm(\omega_{\mu}+\omega_{\nu}-\omega_{l-1}-\omega_{l}).\end{gathered}$$	
killing1889_EQ0106	028	— S|+1	\begin{aligned} & \omega_{1}, \ldots \omega_{6}, \omega_{4}-\omega_{x}, \omega_{6}-\omega_{4}, \omega_{6}-\omega_{5}, \omega_{6}-\omega_{6}, \omega_{4}-\omega_{5}, \\ & -\omega_{4}+\omega_{4}+\omega_{6}, -\omega_{4}+\omega_{5}+\omega_{6}, -\omega_{4}-\omega_{x}+\omega_{4}+\omega_{6}, \\ & -\omega_{4}-\omega_{x}+\omega_{5}+\omega_{6}, -\omega_{4}-\omega_{x}+\omega_{4}+\omega_{5}+\omega_{6}, \\ & -\omega_{1}-\omega_{2}-\omega_{3}+\omega_{4}+\omega_{5}+\omega_{6}, -\omega_{1}-\omega_{2}-\omega_{3}+\omega_{4}+\omega_{5}+2 \omega_{6} . \end{aligned}	$$\begin{aligned}&\omega_{1},\ldots\omega_{6},\ \omega_{4}-\omega_{x},\ \omega_{6}-\omega_{4},\ \omega_{6}-\omega_{5},\ \omega_{6}-\omega_{6},\ \omega_{4}-\omega_{5},\\ &-\omega_{4}+\omega_{4}+\omega_{6},\ -\omega_{4}+\omega_{5}+\omega_{6},\ -\omega_{4}-\omega_{x}+\omega_{4}+\omega_{6},\\ &-\omega_{4}-\omega_{x}+\omega_{5}+\omega_{6},\ -\omega_{4}-\omega_{x}+\omega_{4}+\omega_{5}+\omega_{6},\\ &-\omega_{1}-\omega_{2}-\omega_{3}+\omega_{4}+\omega_{5}+\omega_{6},\ -\omega_{1}-\omega_{2}-\omega_{3}+\omega_{4}+\omega_{5}+2\omega_{6}.\end{aligned}$$	

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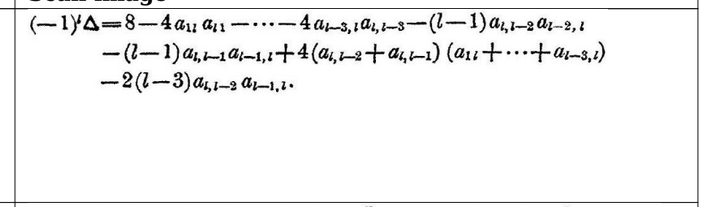
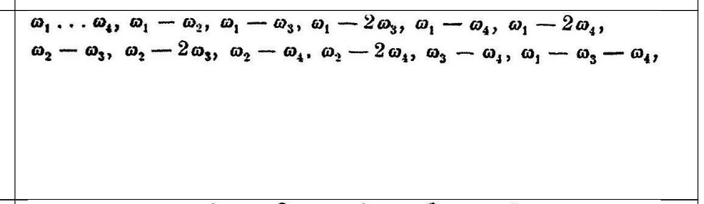
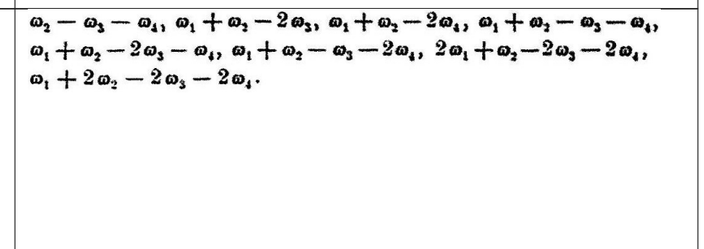
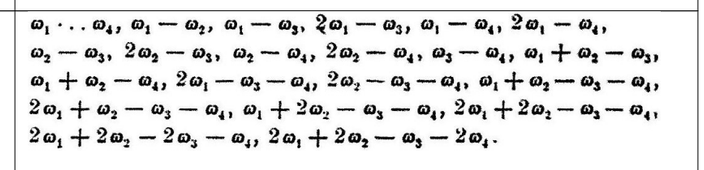
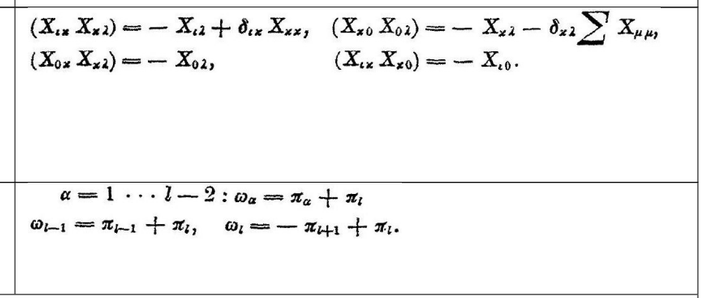
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
killing1889_EQ0107	028	— ● W|+2	\begin{aligned} & \omega_{1} \ldots \omega_{7}, \omega_{4}-\omega_{x}, \omega_{4}-\omega_{5}, \omega_{4}-\omega_{6}, \omega_{5}-\omega_{6}, \omega_{4}-\omega_{7}, \\ & -\omega_{4}+\omega_{5}+\omega_{7},-\omega_{4}+\omega_{6}+\omega_{7},-\omega_{4}-\omega_{x}+\omega_{5}+\omega_{7}, \\ & -\omega_{4}-\omega_{x}+\omega_{6}+\omega_{7},-\omega_{1}-\omega_{x}+\omega_{5}+\omega_{6}+\omega_{7}, \\ & -\omega_{4}-\omega_{x}-\omega_{2}+\omega_{5}+\omega_{6}+\omega_{7},-\omega_{4}-\omega_{x}-\omega_{2}+\omega_{5}+\omega_{6}+2\omega_{7}, \\ & \omega_{1}+\omega_{2}+\omega_{3}+\omega_{4}-\omega_{5}-\omega_{6}-2\omega_{7},\omega_{1}+\cdots+\omega_{4}-2\omega_{5}-\omega_{6}-2\omega_{7}, \\ & \omega_{1}+\cdots+\omega_{4}-\omega_{5}-2\omega_{6}-2\omega_{7}. \end{aligned}	$\omega_1 \ldots \omega_7, \omega_4-\omega_x, \omega_4-\omega_5, \omega_4-\omega_6, \omega_5-\omega_6, \omega_4-\omega_7,$ $-\omega_4+\omega_5+\omega_7, -\omega_4+\omega_6+\omega_7, -\omega_4-\omega_x+\omega_5+\omega_7,$ $-\omega_4-\omega_x+\omega_6+\omega_7, -\omega_1-\omega_x+\omega_5+\omega_6+\omega_7,$ $-\omega_4-\omega_x-\omega_2+\omega_5+\omega_6+\omega_7, -\omega_4-\omega_x-\omega_2+\omega_5+\omega_6+2\omega_7,$ $\omega_1+\omega_2+\omega_3+\omega_4-\omega_5-\omega_6-2\omega_7, \omega_1+\cdots+\omega_4-2\omega_5-\omega_6-2\omega_7,$ $\omega_1+\cdots+\omega_4-\omega_5-2\omega_6-2\omega_7.$	
killing1889_EQ0108	028	— ● C|-6	\begin{gathered} a_{4}=-1, a_{16}=a_{47}=a_{48}=a_{64}=a_{71}=a_{56}=-1, a_{67}=a_{76}=-1, \\ a_{65}=a_{56}=a_{75}=a_{57}=0 . \end{gathered}	$a_4=-1, a_{16}=a_{47}=a_{48}=a_{64}=a_{71}=a_{56}=-1, a_{67}=a_{76}=-1,$ $a_{65}=a_{56}=a_{75}=a_{57}=0.$	
killing1889_EQ0109	029	— ●	\begin{aligned} & \omega_{1} \cdots \omega_{3}, \omega_{4}-\omega_{x}, \omega_{4}-\omega_{6}, \omega_{6}-\omega_{7}, \omega_{4}-\omega_{7}, \omega_{4}-\omega_{8}, \omega_{8}-\omega_{6}-\omega_{5}, \\ & \omega_{4}-\omega_{7}-\omega_{5}, \omega_{4}+\omega_{x}-\omega_{6}-\omega_{5}, \omega_{4}+\omega_{x}-\omega_{7}-\omega_{5}, \\ & \omega_{4}+\omega_{x}-\omega_{6}-\omega_{7}-\omega_{5}, \omega_{4}+\omega_{x}+\omega_{2}-\omega_{6}-\omega_{7}-\omega_{5}, \\ & \omega_{4}+\omega_{x}+\omega_{2}-\omega_{6}-\omega_{7}-2\omega_{8}, \omega_{4}+\omega_{x}+\omega_{2}+\omega_{4}-\omega_{6}-\omega_{7}-2\omega_{5}, \end{aligned}	$\omega_1 \cdots \omega_3, \omega_4-\omega_x, \omega_4-\omega_6, \omega_6-\omega_7, \omega_4-\omega_7, \omega_4-\omega_8, \omega_8-\omega_6-\omega_5,$ $\omega_4-\omega_7-\omega_5, \omega_4+\omega_x-\omega_6-\omega_5, \omega_4+\omega_x-\omega_7-\omega_5,$ $\omega_4+\omega_x-\omega_6-\omega_7-\omega_5, \omega_4+\omega_x+\omega_2-\omega_6-\omega_7-\omega_5,$ $\omega_4+\omega_x+\omega_2-\omega_6-\omega_7-2\omega_8, \omega_4+\omega_x+\omega_2+\omega_4-\omega_6-\omega_7-2\omega_5,$	
killing1889_EQ0114	029	— ● C|-29	\begin{aligned} (-1)^{i} \Delta & =4-2 a_{11} a_{1 i}-2 a_{i 2} a_{2 i}-2 a_{i 2} a_{2 i}-\cdots-2 a_{i-2, l} a_{i, l-2}-(l-1) a_{i, l-1} a_{l-1, l} \\ & \\ & +4 a_{l-1, l}\left(a_{l i}+\cdots+a_{l, l-2}\right) . \end{aligned}	$(-1)^i\Delta = 4-2a_{11}a_{1i}-2a_{i2}a_{2i}-\cdots-2a_{i-2,l}a_{i,l-2}-(l-1)a_{i,l-1}a_{l-1,l}$ $+4a_{l-1,l}\left(a_{li}+\cdots+a_{l,l-2}\right).$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix's reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● X crop overran its region ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

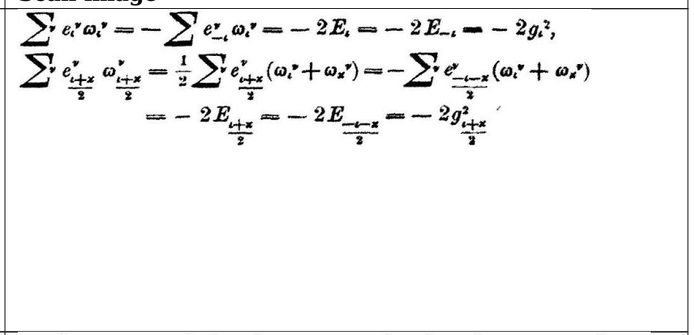
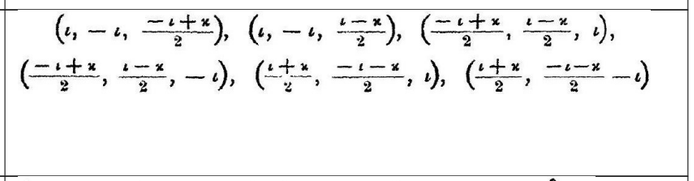
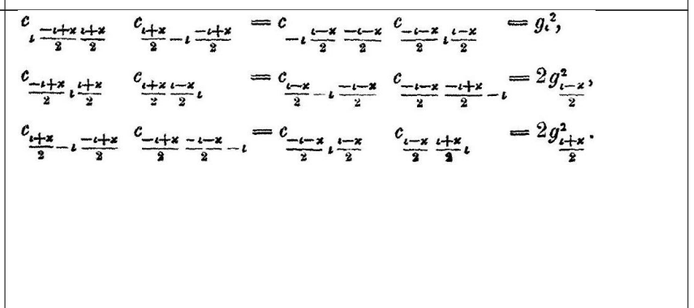
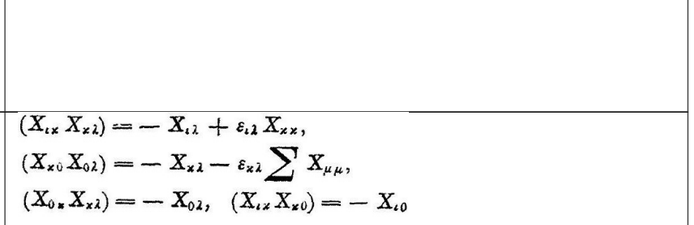
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
killing1889_EQ0119	030	— ● C|-13	\begin{aligned} (-1)^{t} \Delta & =8-4 a_{1 l} a_{l 1}-\cdots-4 a_{l-3, l} a_{l, l-3}-(l-1) a_{l, l-2} a_{l-2, l} \\ & -(l-1) a_{l, l-1} a_{l-1, l}+4\left(a_{l, l-2}+a_{l, l-1}\right)\left(a_{1 l}+\cdots+a_{l-3, l}\right) \\ & -2(l-3) a_{b, l-2} a_{l-1, l} . \end{aligned}	$(-1)^t\Delta = 8 - 4a_{1l}a_{l1} - \cdots - 4a_{l-3,l}a_{l,l-3} - (l-1)a_{l,l-2}a_{l-2,l}$ $- (l-1)a_{l,l-1}a_{l-1,l} + 4\,(a_{l,l-2} + a_{l,l-1})\,(a_{1l} + \cdots + a_{l-3,l})$ $- 2(l-3)a_{b,l-2}a_{l-1,l}.$	
killing1889_EQ0121	030	— ● X|+2	\begin{aligned} & \omega_{1} \ldots \omega_{4}, \omega_{1}-\omega_{2}, \omega_{1}-\omega_{3}, \omega_{1}-2 \omega_{3}, \omega_{1}-\omega_{4}, \omega_{1}-2 \omega_{4}, \\ & \omega_{2}-\omega_{3}, \omega_{2}-2 \omega_{3}, \omega_{2}-\omega_{4}, \omega_{2}-2 \omega_{4}, \omega_{3}-\omega_{4}, \omega_{1}-\omega_{3}-\omega_{4}, \end{aligned}	$\omega_1 \ldots \omega_4, \omega_1 - \omega_2, \omega_1 - \omega_3, \omega_1 - 2\omega_3, \omega_1 - \omega_4, \omega_1 - 2\omega_4,$ $\omega_2 - \omega_3, \omega_2 - 2\omega_3, \omega_2 - \omega_4, \omega_2 - 2\omega_4, \omega_3 - \omega_4, \omega_1 - \omega_3 - \omega_4,$	
killing1889_EQ0122	031	— ● C|-3	\begin{aligned} & \omega_{2}-\omega_{3}-\omega_{4}, \omega_{1}+\omega_{2}-2 \omega_{3}, \omega_{1}+\omega_{2}-2 \omega_{4}, \omega_{1}+\omega_{2}-\omega_{3}-\omega_{4}, \\ & \omega_{1}+\omega_{2}-2 \omega_{3}-\omega_{4}, \omega_{1}+\omega_{2}-\omega_{3}-2 \omega_{4}, 2 \omega_{1}+\omega_{2}-2 \omega_{3}-2 \omega_{4}, \\ & \omega_{1}+2 \omega_{2}-2 \omega_{3}-2 \omega_{4} . \end{aligned}	$\omega_2 - \omega_3 - \omega_4, \omega_1 + \omega_2 - 2\omega_3, \omega_1 + \omega_2 - 2\omega_4, \omega_1 + \omega_2 - \omega_3 - \omega_4,$ $\omega_1 + \omega_2 - 2\omega_3 - \omega_4, \omega_1 + \omega_2 - \omega_3 - 2\omega_4, 2\omega_1 + \omega_2 - 2\omega_3 - 2\omega_4,$ $\omega_1 + 2\omega_2 - 2\omega_3 - 2\omega_4.$	
killing1889_EQ0123	031	— ●	\begin{aligned} & \omega_{1} \cdot \omega_{4}, \omega_{1}-\omega_{2}, \omega_{1}-\omega_{3}, 2 \omega_{1}-\omega_{3}, \omega_{1}-\omega_{4}, 2 \omega_{1}-\omega_{4}, \\ & \omega_{2}-\omega_{3}, 2 \omega_{2}-\omega_{3}, \omega_{2}-\omega_{4}, 2 \omega_{2}-\omega_{4}, \omega_{3}-\omega_{4}, \omega_{1}+\omega_{2}-\omega_{3}, \\ & \omega_{1}+\omega_{2}-\omega_{4}, 2 \omega_{1}-\omega_{3}-\omega_{4}, 2 \omega_{2}-\omega_{3}-\omega_{4}, \omega_{1}+\omega_{2}-\omega_{3}-\omega_{4}, \\ & 2 \omega_{1}+\omega_{2}-\omega_{3}-\omega_{4}, \omega_{1}+2 \omega_{2}-\omega_{3}-\omega_{4}, 2 \omega_{1}+2 \omega_{2}-\omega_{3}-\omega_{4}, \\ & 2 \omega_{1}+2 \omega_{2}-2 \omega_{3}-\omega_{4}, 2 \omega_{1}+2 \omega_{2}-\omega_{3}-2 \omega_{4} . \end{aligned}	$\omega_1 \cdot \omega_4, \omega_1 - \omega_2, \omega_1 - \omega_3, 2\omega_1 - \omega_3, \omega_1 - \omega_4, 2\omega_1 - \omega_4,$ $\omega_2 - \omega_3, 2\omega_2 - \omega_3, \omega_2 - \omega_4, 2\omega_2 - \omega_4, \omega_3 - \omega_4, \omega_1 + \omega_2 - \omega_3,$ $\omega_1 + \omega_2 - \omega_4, 2\omega_1 - \omega_3 - \omega_4, 2\omega_2 - \omega_3 - \omega_4, \omega_1 + \omega_2 - \omega_3 - \omega_4,$ $2\omega_1 + \omega_2 - \omega_3 - \omega_4, \omega_1 + 2\omega_2 - \omega_3 - \omega_4, 2\omega_1 + 2\omega_2 - \omega_3 - \omega_4,$ $2\omega_1 + 2\omega_2 - 2\omega_3 - \omega_4, 2\omega_1 + 2\omega_2 - \omega_3 - 2\omega_4.$	
killing1889_EQ0141	038	— ● C|-9	\begin{array}{ll} \left(X_{i x} X_{x 2}\right)=-X_{i 2}+\delta_{i x} X_{x x}, & \left(X_{x 0} X_{0 i}\right)=-X_{x i}-\delta_{x i} \sum X_{\mu \mu}, \\ \left(X_{0 x} X_{x 2}\right)=-X_{02}, & \left(X_{i x} X_{x 0}\right)=-X_{i 0}. \end{array}	$(X_{ix}X_{x2}) = -X_{i2} + \delta_{ix}X_{xx}, \quad (X_{x0}X_{0i}) = -X_{xi} - \delta_{xi}\sum X_{\mu\mu},$ $(X_{0x}X_{x2}) = -X_{02}, \qquad (X_{ix}X_{x0}) = -X_{i0}.$	
killing1889_EQ0143	038	— ● C|-3	\begin{aligned} \alpha & =1 \cdots l-2: \omega_{\alpha}=\pi_{\alpha}+\pi_{l} \\ \omega_{l-1} & =\pi_{l-1}+\pi_{l}, \quad \omega_{l}=-\pi_{l+1}+\pi_{l} . \end{aligned}	$\alpha = 1 \cdots l-2 : \omega_\alpha = \pi_\alpha + \pi_l$ $\omega_{l-1} = \pi_{l-1} + \pi_l, \quad \omega_l = -\pi_{l+1} + \pi_l.$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix's reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● X crop overran its region ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

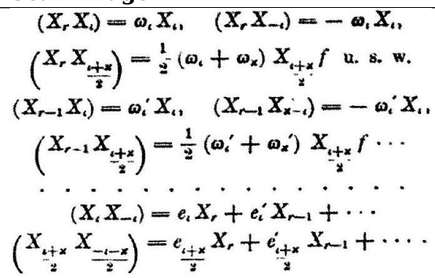
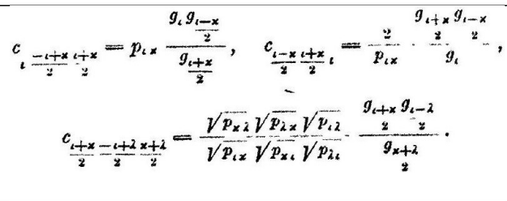
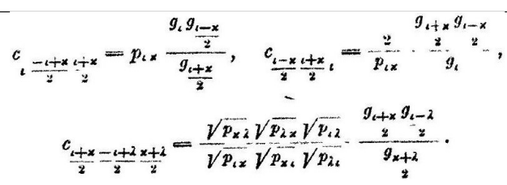
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
killing1889_ EQ0148 (2)	039	— ● C|-26	\begin{gathered} \sum i e_{i}^{v} \omega_{i}^{v}=-\sum e_{-i}^{v} \omega_{i}^{v}=-2 E_{i}=-2 E_{-i}--2 g_{i}^{2}, \\ \sum e_{\frac{i+x}{2}}^{v} \omega_{\frac{i+x}{2}}^{v}=\frac{1}{2} \sum e_{\frac{4+x}{2}}^{v}\left(\omega_{i}^{2}+\omega_{x}^{2}\right)=-\sum \frac{e^{v}-x}{2}\left(\omega_{i}^{v}+\omega_{x}^{\prime}\right) \\ =-2 E_{\frac{4+x}{2}}=-2 E_{\frac{-t-x}{2}}=-2 g_{\frac{i+x}{2}}^{2} \end{gathered}	$$\sum i e_i^v \omega_i^v = -\sum e_{-i}^v \omega_i^v = -2E_i = -2E_{-i} - -2g_i^2,$$ $$\sum e_{\frac{i+x}{2}}^v \omega_{\frac{i+x}{2}}^v = \frac{1}{2}\sum e_{\frac{4+x}{2}}^v \left(\omega_i^2+\omega_x^2\right) = -\sum \frac{e^v - x}{2}\left(\omega_i^v + \omega_x'\right)$$ $$= -2E_{\frac{4+x}{2}} = -2E_{\frac{-t-x}{2}} = -2g_{\frac{i+x}{2}}^2$$	
killing1889_EQ0151	040	— ● X|+24	\begin{array}{rl} \quad\left(i,-i, \frac{-i+x}{2}\right), & \left(i,-i, \frac{i-x}{2}\right), \\ \left(\frac{-i+x}{2}, \frac{i-x}{2},-i\right), & \left(\frac{i+x}{2}, \frac{-i-x}{2}, i\right), \\ 2 & i+x \end{array}	$$\begin{array}{rl} \left(i,-i,\frac{-i+x}{2}\right), & \left(i,-i,\frac{i-x}{2}\right), \\ \left(\frac{-i+x}{2},\frac{i-x}{2},-i\right), & \left(\frac{i+x}{2},\frac{-i-x}{2},i\right), \\ 2 & i+x \end{array}$$	
killing1889_EQ0152	040	— ●	\begin{aligned} & c \frac{-t+x i+x}{2} \quad c_{\frac{t+x}{2}-t-t+x}^{2}=c_{-t} \frac{t-x}{2} \frac{-t-x}{2} \frac{c-t-x}{2} \iota \frac{t-x}{2}=g_{t}^{2}, \\ & \frac{c_{-t+x}}{2} \iota \frac{t+x}{2} \quad c_{\frac{t+x}{2} \frac{t-x}{2} \iota}=c_{\frac{t-x}{2}-\iota \frac{-t-x}{2}} \frac{c_{-t-x}}{2} \frac{-t+x}{2}-\iota=2 g_{t-x}^{2}, \\ & \frac{c_{t+x}}{2}, \frac{-t+x}{2} \quad \frac{c-t+x}{2} \frac{-t-x}{2}=\frac{c_{-t-x}}{2}, \frac{t-x}{2} \quad c_{\frac{t-x}{2} \frac{t+x}{2} \iota}^{2}=2 g_{\frac{t+x}{2}}^{2} . \end{aligned}	$$c\frac{-t+xi+x}{2} \quad c^2_{\frac{t+x}{2}-t-t+x} = c_{-t}\frac{t-x}{2}\frac{-t-x}{2}\frac{c-t-x}{2}\iota\frac{t-x}{2} = g_t^2,$$ $$\frac{c_{-t+x}}{2}\iota\frac{t+x}{2} \quad c_{\frac{t+x}{2}\frac{t-x}{2}\iota} = c_{\frac{t-x}{2}-\iota\frac{-t-x}{2}}\frac{c_{-t-x}}{2}\frac{-t+x}{2} - \iota = 2g^2_{t-x},$$ $$\frac{c_{t+x}}{2}, \frac{-t+x}{2} \quad \frac{c-t+x}{2}\frac{-t-x}{2} = \frac{c_{-t-x}}{2}, \frac{t-x}{2} \quad c^2_{\frac{t-x}{2}\frac{t+x}{2}\iota} = 2g^2_{\frac{t+x}{2}}.$$	
killing1889_EQ0161	042	— ● C|-6	\begin{aligned} & \left(X_{i x} X_{x \lambda}\right)=-X_{i \lambda}+\varepsilon_{i \lambda} X_{x x}, \\ & \left(X_{x 0} X_{02}\right)=-X_{x \lambda}-\varepsilon_{x \lambda} \sum X_{\mu \mu}, \\ & \left(X_{0} X_{x \lambda}\right)=-X_{0 \lambda}, \quad\left(X_{i \lambda} X_{x 0}\right)=-X_{i 0} \end{aligned}	$$\begin{aligned} & \left(X_{ix}X_{x\lambda}\right) = -X_{i\lambda} + \varepsilon_{i\lambda}X_{xx}, \\ & \left(X_{x0}X_{02}\right) = -X_{x\lambda} - \varepsilon_{x\lambda}\sum X_{\mu\mu}, \\ & \left(X_0 X_{x\lambda}\right) = -X_{0\lambda}, \quad \left(X_{i\lambda}X_{x0}\right) = -X_{i0} \end{aligned}$$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix's reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● X crop overran its region ● not measured

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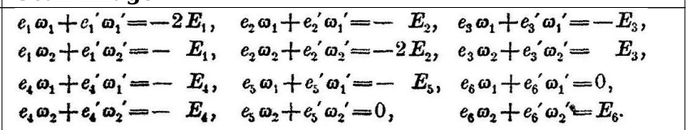
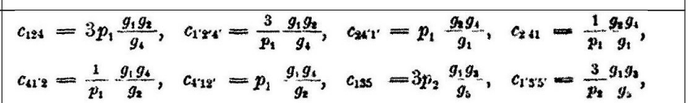
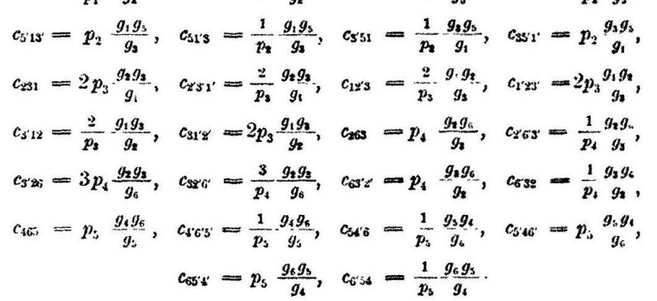
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
killing1889_EQ0164	043	— W|-2	\begin{gathered} \left(X_{r} X_{i}\right)=\omega_{i} X_{i} \quad\left(X_{r} X_{-i}\right)=-\omega_{i} X_{i} \\ \left(X_{r} X_{i+x}\right)=\frac{1}{2}\left(\omega_{i}+\omega_{x}\right) X_{i+x} f \text { u. s. w. } \\ \left(X_{r-1} X_{i}\right)=\omega_{i}^{\prime} X_{i} \quad\left(X_{r-1} X_{x-i}\right)=-\omega_{i}^{\prime} X_{i} \\ \left(X_{r-1} X_{i+x}\right)=\frac{1}{2}\left(\omega_{i}^{\prime}+\omega_{x}^{\prime}\right) X_{i+x} f \cdots \\ \left(X_{i} X_{-i}\right)=e_{i} X_{i}+e_{i}^{\prime} X_{r-1}+\cdots \\ \left(X_{i+x} X_{i-x}^{\prime}\right)=e_{\frac{i+x}{2}} X_{r}+e_{i+x}^{\prime} X_{r-1}+\cdots \end{gathered}	$$\begin{gathered}(X_rX_i)=\omega_iX_i \quad (X_rX_{-i})=-\omega_iX_i\\(X_rX_{i+x})=\frac{1}{2}(\omega_i+\omega_x)X_{i+x}f \text{ u. s. w.}\\(X_{r-1}X_i)=\omega_i'X_i \quad (X_{r-1}X_{x-i})=-\omega_i'X_i,\\(X_{r-1}X_{i+x})=\frac{1}{2}(\omega_i'+\omega_x')X_{i+x}f\cdots\\(X_iX_{-i})=e_iX_r+e_i'X_{r-1}+\cdots\\(X_{i+x}X_{i-x}')=e_{\frac{i+x}{2}}X_r+e_{i+x}'X_{r-1}+\cdots\end{gathered}$$	
killing1889_EQ0168	043	— C|+154	—	—	
killing1889_EQ0169	043	—	\begin{aligned} & c_{i-i+x i+x}=p_{i x} \frac{g_{i} g_{i-x}}{g_{\frac{i+x}{2}}}, \quad c_{\frac{i-x}{2} \frac{i+x}{2} \frac{i}{2}}=\frac{2}{p_{i x}}=\frac{g_{i+x} g_{i-x}}{g_{i}}, \\ & c_{\frac{i+x-i+2 x+2}{2}}=\frac{\sqrt{p_{x i}} \sqrt{p_{i x}} \sqrt{p_{i, 2}}}{\sqrt{p_{i x}} \sqrt{p_{x i}} \sqrt{p_{i i}}} \frac{g_{i+x} g_{i-2}}{\frac{2}{g_{x+2}}}. \end{aligned}	$$\begin{aligned}&c_{i-i+xi+x}=p_{ix}\frac{g_ig_{i-x}}{\frac{g_{i+x}}{2}}, \quad c_{\frac{i-x}{2}\frac{i+x}{2}\frac{i}{2}}=\frac{2}{p_{ix}}=\frac{g_{i+x}g_{i-x}}{g_i},\\&c_{\frac{i+x-i+2x+2}{2}}=\frac{\sqrt{p_{xi}}\sqrt{p_{ix}}\sqrt{p_{i,2}}}{\sqrt{p_{ix}}\sqrt{p_{xi}}\sqrt{p_{ii}}}\frac{g_{i+x}g_{i-2}}{\frac{2}{g_{x+2}}}.\end{aligned}$$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix's reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● X crop overran its region ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

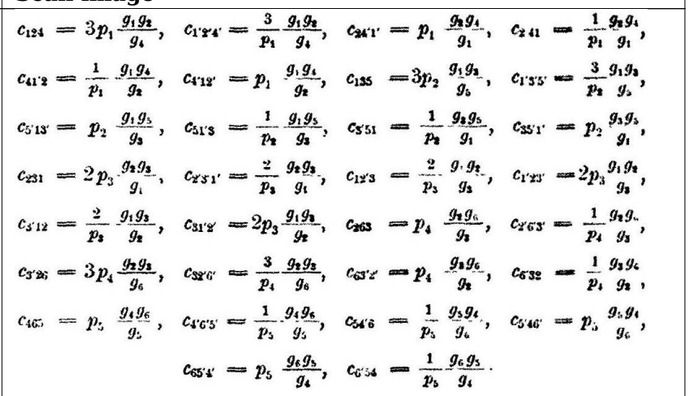
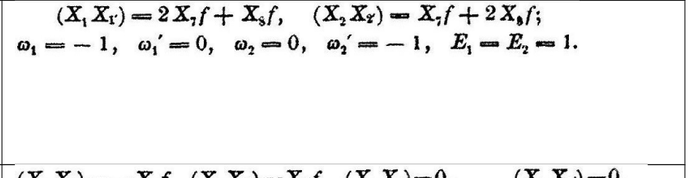
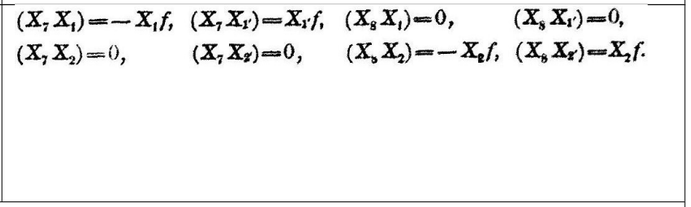
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
killing1889_EQ0172	044	— ● C|-15	\begin{array}{lll} e_{1} \omega_{1}+e_{1}^{\prime} \omega_{1}^{\prime}=-2 E_{1}, & e_{2} \omega_{1}+e_{2}^{\prime} \omega_{1}^{\prime}=-E_{2}, & e_{3} \omega_{1}+e_{3}^{\prime} \omega_{1}^{\prime}=-E_{3}, \\ e_{1} \omega_{2}+e_{1}^{\prime} \omega_{2}^{\prime}=-E_{1}, & e_{2} \omega_{2}+e_{2}^{\prime} \omega_{2}^{\prime}=-2 E_{2}, & e_{3} \omega_{2}+e_{3}^{\prime} \omega_{2}^{\prime}=E_{3}, \\ e_{4} \omega_{1}+e_{4}^{\prime} \omega_{1}^{\prime}=-E_{4}, & e_{5} \omega_{1}+e_{5}^{\prime} \omega_{1}^{\prime}=-E_{5}, & e_{6} \omega_{1}+e_{6}^{\prime} \omega_{1}^{\prime}=0, \\ e_{4} \omega_{2}+e_{4}^{\prime} \omega_{2}^{\prime}=-E_{4}, & e_{5} \omega_{2}+e_{5}^{\prime} \omega_{2}^{\prime}=0, & e_{6} \omega_{2}+e_{6}^{\prime} \omega_{2}^{\prime}=E_{6} . \end{array}	$$\begin{array}{lll} e_{1} \omega_{1}+e_{1}^{\prime} \omega_{1}^{\prime}=-2 E_{1}, & e_{2} \omega_{1}+e_{2}^{\prime} \omega_{1}^{\prime}=-E_{2}, & e_{3} \omega_{1}+e_{3}^{\prime} \omega_{1}^{\prime}=-E_{3}, \\ e_{1} \omega_{2}+e_{1}^{\prime} \omega_{2}^{\prime}=-E_{1}, & e_{2} \omega_{2}+e_{2}^{\prime} \omega_{2}^{\prime}=-2 E_{2}, & e_{3} \omega_{2}+e_{3}^{\prime} \omega_{2}^{\prime}=E_{3}, \\ e_{4} \omega_{1}+e_{4}^{\prime} \omega_{1}^{\prime}=-E_{4}, & e_{5} \omega_{1}+e_{5}^{\prime} \omega_{1}^{\prime}=-E_{5}, & e_{6} \omega_{1}+e_{6}^{\prime} \omega_{1}^{\prime}=0, \\ e_{4} \omega_{2}+e_{4}^{\prime} \omega_{2}^{\prime}=-E_{4}, & e_{5} \omega_{2}+e_{5}^{\prime} \omega_{2}^{\prime}=0, & e_{6} \omega_{2}+e_{6}^{\prime} \omega_{2}^{\prime}=E_{6}. \end{array}$$	
killing1889_EQ0173	045	— ● C|-14	\begin{aligned} \frac{e_{2}}{E_{3}}=\frac{e_{1}}{E_{1}} -\frac{e_{2}}{E_{2}}, & \frac{e_{4}}{E_{1}}=\frac{1}{3} \left(\frac{e_{1}}{E_{1}}+\frac{e_{2}}{E_{2}}\right), \quad \frac{e_{3}}{E_{5}}=\frac{1}{3}\left(\frac{2 e_{1}}{E_{1}}-\frac{e_{2}}{E_{2}}\right) . \\ & \frac{e_{6}}{E_{6}}=\frac{1}{3}\left(\frac{e_{1}}{E_{1}}-2 \frac{e_{6}}{E_{2}}\right) \end{aligned}	$$\begin{aligned} \frac{e_{2}}{E_{3}}=\frac{e_{1}}{E_{1}}-\frac{e_{2}}{E_{2}}, & \frac{e_{4}}{E_{1}}=\frac{1}{3}\left(\frac{e_{1}}{E_{1}}+\frac{e_{2}}{E_{2}}\right), \quad \frac{e_{3}}{E_{5}}=\frac{1}{3}\left(\frac{2 e_{1}}{E_{1}}-\frac{e_{2}}{E_{2}}\right) . \\ & \frac{e_{6}}{E_{6}}=\frac{1}{3}\left(\frac{e_{1}}{E_{1}}-2 \frac{e_{6}}{E_{2}}\right) \end{aligned}$$	
killing1889_EQ0174	045	— ● C|+454	—	—	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix's reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● X crop overran its region ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

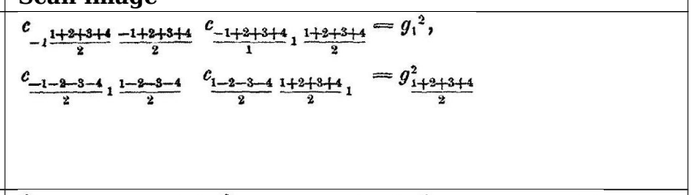
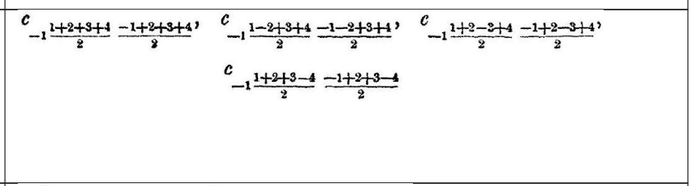
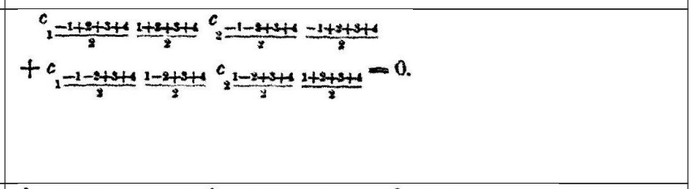
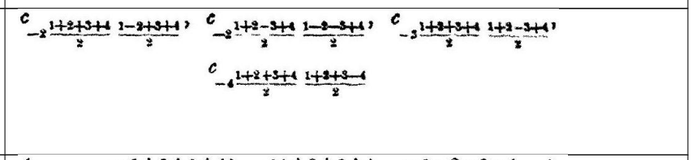
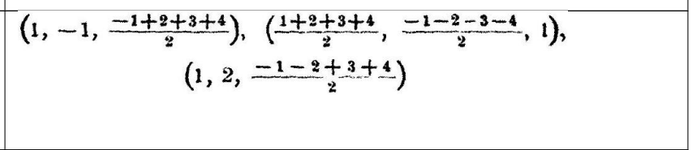
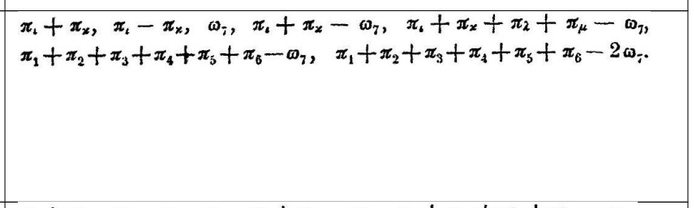
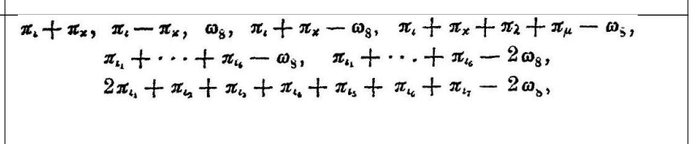
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
killing1889_E00175	045	— ●	\begin{aligned} & c_{124}=3 p_{1} \frac{g_{1} g_{2}}{g_{4}}, \quad c_{124}=\frac{3}{p_{1}} \frac{g_{1} g_{2}}{g_{4}}, \quad c_{241}=p_{1} \frac{g_{2} g_{4}}{g_{1}}, \quad c_{241}=\frac{1}{p_{1}} \frac{g_{2} g_{4}}{g_{1}}, \\ & c_{422}=\frac{1}{p_{1}} \frac{g_{1} g_{4}}{g_{2}}, \quad c_{412}=p_{1} \frac{g_{1} g_{1}}{g_{2}}, \quad c_{135}=3 p_{2} \frac{g_{1} g_{3}}{g_{3}}, \quad c_{155}=\frac{3}{p_{2}} \frac{g_{1} g_{3}}{g_{3}}, \\ & c_{515}=p_{2} \frac{g_{1} g_{3}}{g_{3}}, \quad c_{515}=\frac{1}{p_{2}} \frac{g_{1} g_{5}}{g_{3}}, \quad c_{551}=\frac{1}{p_{2}} \frac{g_{3} g_{5}}{g_{1}}, \quad c_{551}=p_{2} \frac{g_{3} g_{5}}{g_{1}}, \\ & c_{231}=2 p_{3} \frac{g_{2} g_{3}}{g_{1}}, \quad c_{251}=\frac{2}{p_{3}} g_{2} g_{3}, \quad c_{12^{\prime} 3}=\frac{2}{p_{3}} \quad g_{3} \quad g_{2}, \quad c_{123}=2 p_{3} \quad g_{1} g_{2}, \\ & c_{s 12}=\frac{2}{p_{s}} \frac{g_{1} g_{3}}{g_{2}}, \quad c_{312}=2 p_{3} \frac{g_{1} g_{3}}{g_{2}}, \quad c_{263}=p_{4} \frac{g_{2} g_{6}}{g_{3}}, \quad c_{265}=\frac{1}{p_{4}} \frac{g_{2} g_{3}}{g_{3}}, \\ & c_{\zeta^{2} \zeta^{3}}=3 p_{4} \frac{g_{2} g_{3}}{g_{6}}, \quad c_{3 \zeta^{\prime}}=\frac{3}{p_{4}} \frac{g_{2} g_{3}}{g_{6}}, \quad c_{632}=p_{4} \frac{g_{2} g_{6}}{g_{2}}, \quad c_{632}=\frac{1}{p_{1}} \frac{g_{2} g_{6}}{g_{2}}, \\ & c_{46}=p_{3} \frac{g_{4} g_{6}}{g_{3}}, c_{463}=\frac{1}{p_{3}} \frac{g_{4} g_{5}}{g_{5}}, c_{56}=\frac{1}{p_{3}} \frac{g_{3} g_{4}}{g_{6}}, c_{546}=p_{5} \frac{g_{5} g_{6}}{g_{6}}, \\ & c_{654}=p_{5} \frac{g_{6} g_{3}}{g_{4}}, \quad c_{634}=\frac{1}{p_{5}} \frac{g_{6} g_{5}}{g_{4}} . \end{aligned}	$c_{124}=3p_1\frac{g_1g_2}{g_4}$, $c_{124}=\frac{3}{p_1}\frac{g_1g_2}{g_4}$, $c_{241}=p_1\frac{g_2g_4}{g_1}$, $c_{241}=\frac{1}{p_1}\frac{g_2g_4}{g_1}$, $c_{422}=\frac{1}{p_1}\frac{g_1g_4}{g_2}$, $c_{412}=p_1\frac{g_1g_1}{g_2}$, $c_{135}=3p_2\frac{g_1g_3}{g_3}$, $c_{155}=\frac{3}{p_2}\frac{g_1g_3}{g_3}$, $c_{515}=p_2\frac{g_1g_3}{g_3}$, $c_{515}=\frac{1}{p_2}\frac{g_1g_5}{g_3}$, $c_{551}=\frac{1}{p_2}\frac{g_3g_5}{g_1}$, $c_{551}=p_2\frac{g_3g_5}{g_1}$, $c_{231}=2p_3\frac{g_2g_3}{g_1}$, $c_{251}=\frac{2}{p_3}g_2g_3$, $c_{12'3}=\frac{2}{p_3}\quad g_3\quad g_2$, $c_{123}=2p_3\quad g_1g_2$, $c_{s12}=\frac{2}{p_s}\frac{g_1g_3}{g_2}$, $c_{312}=2p_3\frac{g_1g_3}{g_2}$, $c_{263}=p_4\frac{g_2g_6}{g_3}$, $c_{265}=\frac{1}{p_4}\frac{g_2g_3}{g_3}$, $c_{\zeta^2\zeta^3}=3p_4\frac{g_2g_3}{g_6}$, $c_{3\zeta'}=\frac{3}{p_4}\frac{g_2g_3}{g_6}$, $c_{632}=p_4\frac{g_2g_6}{g_2}$, $c_{632}=\frac{1}{p_1}\frac{g_2g_6}{g_2}$, $c_{46}=p_3\frac{g_4g_6}{g_3}$, $c_{463}=\frac{1}{p_3}\frac{g_4g_5}{g_5}$, $c_{56}=\frac{1}{p_3}\frac{g_3g_4}{g_6}$, $c_{546}=p_5\frac{g_5g_6}{g_6}$, $c_{654}=p_5\frac{g_6g_3}{g_4}$, $c_{634}=\frac{1}{p_5}\frac{g_6g_5}{g_4}$.	
killing1889_E00177	045	— ● C|-9	\begin{gathered} \left(X_{1} X_{1}^{\prime}\right)=2 X_{7} f+X_{8} f, \quad\left(X_{2} X_{7}\right)=X_{7} f+2 X_{3} f ; \\ \omega_{1}=-1, \quad \omega_{1}^{\prime}=0, \quad \omega_{2}=0, \quad \omega_{2}^{\prime}=-1, \quad E_{1}=E_{2}=1 . \end{gathered}	$(X_1X_1')=2X_7f+X_8f, \quad (X_2X_7)=X_7f+2X_3f;$ $\omega_1=-1, \quad \omega_1'=0, \quad \omega_2=0, \quad \omega_2'=-1, \quad E_1=E_2=1.$	
killing1889_E00179	045	— ● C|-14	\begin{array}{lll} \left(X_{7} X_{1}\right)=-X_{1} f, & \left(X_{7} X_{1}\right)=X_{1} f, & \left(X_{8} X_{1}\right)=X_{1} f, \\ \left(X_{8} X_{1}\right)=0, & \left(X_{7} X_{2}\right)=0, & \left(X_{7} X_{2}\right)=0, \\ \left(X_{8} X_{2}\right)=-X_{8} f, & \left(X_{8} X_{2}\right)=X_{2} f, & \left(X_{8} X_{2}\right)=X_{2} f . \end{array}	$(X_7X_1)=-X_1f, \quad (X_7X_1)=X_1f, \quad (X_8X_1)=X_1f, \quad (X_8X_1)=0, \quad (X_7X_2)=0, \quad (X_7X_2)=0, \quad (X_8X_2)=-X_8f, \quad (X_8X_2)=X_2f.$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix's reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● X crop overran its region ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
killing1889_ EQ0182 (1)	046	— ●	\begin{aligned} & c_{-1} \frac{1+2+3+4}{2} \frac{-1+2+3+4}{2} c_{-1+2+3+4} \frac{1+2+3+4}{2}=g_{1}{ }^{2}, \\ & c_{-1-2-3-4} \frac{1-2-3-4}{2} c_{\frac{1-2-3-4}{2} \frac{1+2+3+4}{2}}=g_{\frac{1+2+3+4}{2}}^{2} \end{aligned}	$c_{-1}\frac{1+2+3+4}{2}\frac{-1+2+3+4}{2}c_{-1+2+3+4}\frac{1+2+3+4}{2}=g_1{}^2,$ $c_{-1-2-3-4}\frac{1-2-3-4}{2}c_{\frac{1-2-3-4}{2}\frac{1+2+3+4}{2}}=g^2_{\frac{1+2+3+4}{2}}$	
killing1889_ EQ0183 (a)	046	— ● C|-24	\begin{gathered} c_{-1} \frac{1+2+3+4}{2} \frac{-1+2+3+4}{2}, \quad c_{-1} \frac{1-2+3+4}{2} \frac{-1-2+3+4}{2}, \quad c_{-1} \frac{1+2-3+4}{2} \frac{-1+2-3+4}{2}, \\ c_{-1} \frac{1+2+3-4}{2} \frac{-1+2+3-4}{2} \end{gathered}	$c_{-1}\frac{1+2+3+4}{2}\frac{-1+2+3+4}{2}, \quad c_{-1}\frac{1-2+3+4}{2}\frac{-1-2+3+4}{2}, \quad c_{-1}\frac{1+2-3+4}{2}\frac{-1+2-3+4}{2},$ $c_{-1}\frac{1+2+3-4}{2}\frac{-1+2+3-4}{2}$	
killing1889_ EQ0186 (3)	047	— ● C|-37	\begin{gathered} c_{1} \frac{-1+2+3+4}{2} \frac{1+2+3+4}{2} c_{2} \frac{-1-2+4+4}{2} \frac{-1+2+3+4}{2} \\ +c_{1} \frac{-1-2+4+1}{2} \frac{1-2+3+1}{2} c_{2} \frac{1-2+3+1}{2} \frac{1+2+3+4}{2}=0 . \end{gathered}	$c_1\frac{-1+2+3+4}{2}\frac{1+2+3+4}{2}c_2\frac{-1-2+4+4}{2}\frac{-1+2+3+4}{2}$ $+c_1\frac{-1-2+4+1}{2}\frac{1-2+3+1}{2}c_2\frac{1-2+3+1}{2}\frac{1+2+3+4}{2}=0.$	
killing1889_EQ0187	047	— ● W|-1	\begin{gathered} C \frac{1+2+3+4}{2} \frac{1-2+3+4}{2}{ } , C{ }_{-2} \frac{1+2-3+4}{2} \frac{1-2-3+4}{2}, C \\ C-\frac{1+2+3+1}{2} \frac{1+2+3+3+1}{2} \end{gathered}	$C\frac{1+2+3+4}{2}\frac{1-2+3+4}{2}, C_{-2}\frac{1+2-3+4}{2}\frac{1-2-3+4}{2}, C$ $C-\frac{1+2+3+1}{2}\frac{1+2+3+3+1}{2}$	
killing1889_EQ0190	047	— ● X|+9	\begin{gathered} \left(1,-1, \frac{-1+2+3+4}{2}\right), \left(\frac{1+2+3+4}{2}, \frac{-1-2-3-4}{2}, 1\right), \\ \left(1,2, \frac{-1-2+3+4}{2}\right) \end{gathered}	$\left(1,-1,\frac{-1+2+3+4}{2}\right), \left(\frac{1+2+3+4}{2},\frac{-1-2-3-4}{2},1\right),$ $\left(1,2,\frac{-1-2+3+4}{2}\right)$	
killing1889_EQ0193	048	— ● W|+0	\begin{aligned} & \pi_{1}+\pi_{x}, \quad \pi_{i}-\pi_{x}, \quad \omega_{7}, \quad \pi_{i}+\pi_{x}-\omega_{7}, \quad \pi_{i}+\pi_{x}+\pi_{\lambda}+\pi_{\mu}-\omega_{7}, \\ & \pi_{1}+\pi_{2}+\pi_{3}+\pi_{4}+\pi_{5}+\pi_{6}-\omega_{7}, \quad \pi_{1}+\pi_{2}+\pi_{3}+\pi_{4}+\pi_{5}+\pi_{6}-2\omega_{7}. \end{aligned}	$\pi_1+\pi_x,\quad \pi_i-\pi_x,\quad \omega_7,\quad \pi_i+\pi_x-\omega_7,\quad \pi_i+\pi_x+\pi_\lambda+\pi_\mu-\omega_7,$ $\pi_1+\pi_2+\pi_3+\pi_4+\pi_5+\pi_6-\omega_7,\quad \pi_1+\pi_2+\pi_3+\pi_4+\pi_5+\pi_6-2\omega_7.$	
killing1889_EQ0194	048	— ● C|-22	\begin{gathered} \pi_{i}+\pi_{x}, \pi_{i}-\pi_{x}, \omega_{8}, \pi_{i}+\pi_{x}-\omega_{8}, \pi_{i}+\pi_{x}+\pi_{2}+\pi_{\mu}-\omega_{5}, \\ \pi_{i_{1}}+\cdots+\pi_{i_{2}}-\omega_{8}, \pi_{i_{1}}+\cdots+\pi_{i_{6}}-2\omega_{8}, \\ 2\pi_{i_{4}}+\pi_{i_{2}}+\pi_{i_{2}}+\pi_{i_{4}}+\pi_{i_{3}}+\pi_{i_{6}}+\pi_{i_{7}}-2\omega_{8}, \end{gathered}	$\pi_i+\pi_x, \pi_i-\pi_x, \omega_8, \pi_i+\pi_x-\omega_8, \pi_i+\pi_x+\pi_2+\pi_\mu-\omega_5,$ $\pi_{i_1}+\cdots+\pi_{i_2}-\omega_8, \pi_{i_1}+\cdots+\pi_{i_6}-2\omega_8,$ $2\pi_{i_4}+\pi_{i_2}+\pi_{i_2}+\pi_{i_4}+\pi_{i_3}+\pi_{i_6}+\pi_{i_7}-2\omega_8,$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix's reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● X crop overran its region ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.